Politechnika Białostocka

ESCUELA POLITÉCNICA SUPERIOR DE CÓRDOBA
Universidad de Córdoba

BACHELOR´S THESIS

Mechanical Engineering Degree

Structural and Mechanical Characterization of Samarium-doped Glass-ionomer Cements

Author
Marco Antonio Rodríguez Vigara

Supervisors:
Magdalena Łępicka
Joost van Duijn

Declaration of authorship of the Final Degree Project

I, Marco Antonio Rodríguez Vigara, holder of National Identity Document (DNI) number 20953641C, student of the Bachelor's Degree in Mechanical Engineering, hereby declare that I am the author of, and assume responsibility for the originality of, the Final Degree Project entitled:

“Structural and Mechanical Characterization of Samarium-doped Glass-ionomer Cements”

Originality is understood in the sense that no sources have been used without being properly cited.

In Bialystok, on 28 May 2026.

Abstract

Glass-ionomer cements are widely used in dentistry because they can bond chemically to dental tissues, release fluoride and show acceptable biocompatibility. However, their mechanical limitations still restrict their use in areas exposed to higher loads. For this reason, the modification of glass-ionomer cement formulations remains an important field of study.

The aim of this work was to evaluate the effect of samarium oxide incorporation on the structural, mechanical and surface properties of experimental glass-ionomer cements. Four fluoroaluminosilicate glass compositions were prepared, using 0, 0.5, 1 and 5 mol% Sm_2O_3 as a partial replacement for SiO_2. Then the experimental cements were characterised by compressive strength testing, surface roughness measurements, Vickers microhardness testing and scanning electron microscopy. In addition, hydrochloric acid pretreatment of the glass powder was explored as a complementary step after the initial mechanical results.

The results showed that Sm_2O_3 incorporation had a limited effect on compressive strength, with no clear statistically significant improvement between the different compositions. However, the surface-related properties were more clearly affected by the presence of samarium. The highest samarium content showed the greatest microhardness values, although it also presented a tendency towards higher surface roughness. Therefore, the effect of samarium was not simply positive or negative, but depended on the property considered and on the concentration used.

Overall, this study suggests that samarium oxide can act as an active modifier of experimental glass-ionomer cements, mainly affecting their surface behaviour rather than their bulk compressive strength. The Sm5 formulation appears to be the most promising from the point of view of microhardness, but further optimisation is needed to improve compressive strength and control surface roughness before considering possible dental applications.

Key words: glass-ionomer cements, mechanical behavior, compressive strength, surface microhardness, microstructural characterization, roughness, fracture analysis, samarium.

Contents

List of figures

List of tables

1. Introduction

Glass-ionomer cements, also known as GICs, are used in dentistry as restorative materials [1]. Their setting is based on an acid-base reaction between a polyacid and a reactive glass powder in the presence of water [1]. These cements appeared as an alternative to older dental silicate cements [2]. These older cements had problems related to fragility, poor adhesion and clinical sensitivity [2]. GICs are useful because they can adhere chemically to dental tissues, release fluoride and show acceptable biocompatibility [3]. For this reason, they are used in restorations, bases, liners, fissure sealants and orthodontic bonding [1].

Their final behaviour depends largely on what happens during setting [1], [4]. In conventional GICs, the most of the powder is calcium fluoroaluminosilicate glass, while the liquid contains polyacrylic acid and water [1], [4]. When both parts are mixed, the acid begins to react with the outer surface of the glass particles [1]. Then calcium and aluminium ions are released into the liquid phase and they progressively link the polymer chains, so the paste starts to harden and become a solid cement [1], [4]. However, the reaction does not consume the whole glass phase [1]. Some particles remain inside the matrix and are known as *unreacted particles* [1]. This remaining glass contributes as a reinforcement for the material [1]. It should be noted that the cement also keeps evolving after the first hardening stage, since maturation can still affect properties such as strength [5].

Despite their advantages, conventional glass-ionomer cements still have certain mechanical limitations [6], [7]. Some of them are their lower fracture resistance, toughness and wear resistance compared with those of natural dental tissues, and their behaviour can also be affected by moisture [6], [8]. Consequently, their use is limited to areas where the restoration must not support high mechanical loads [6], [7]. To improve their properties, many modifications have been studied such as fibres, metal particles, nanoparticles, bioactive glasses, antimicrobial additives and resin-modified systems [8], [9], [10], [11]. Some of these strategies can improve certain properties, but they can also cause problems, for example greater brittleness, lower fluoride release and poorer mechanical response [8], [9], [12].

Samarium is a lanthanide that has proven to have interesting applications in different biomedical materials such as hydroxyapatite, coatings, bioactive glasses and some dental cement-like systems [13], [14], [15], [16]. Previous studies have reported that this element have effects related to biocompatibility, antibacterial behaviour and bioactive response [10], [13].

In this project, experimental GICs were prepared using fluoroaluminosilicate glasses where SiO_2 was partially replaced by 0.5, 1 and 5 mol% of Sm_2O_3. The prepared samples were analysed by compressive strength testing, surface roughness measurements, microhardness testing and scanning electron microscopy. The aim was to evaluate whether the incorporation of Sm_2O_3 produces positive, negative or almost neutral changes in the mechanical, surface and microstructural behaviour of the cement.

2. Literature Review

2.1. Glass-ionomer cements: terminology, development, and general characteristics

Glass-ionomer cements (GICs) are acid-base materials obtained from the reaction of weak polymeric acids with basic glass powders in water. After setting, some of the glass remains unreacted within the solid matrix and acts as a reinforcing phase in the final material [1]. Although the widely used term is "glass-ionomer cement", based strictly on the acid they are composed of the most precise term is "glass polyalkenoate cement" according to the International Organization for Standardization, ISO [1]. Further information about its components will be explained in detail later.

In the late 1960s, with the purpose of substituting dental silicate cements, Alan Wilson and Brian Kent discovered Glass-ionomers at the Laboratory of the Government Chemist, London [2], [4]. Wilson and his collaborators were trying to find a substitute to dental silicate cements because these materials had issues with fragility, erosion produced by acids, poor adhesion to the tooth and gingival sensitivity. After investigating, they concluded that the most suitable solution was to replace the phosphoric acid with a less aggressive acid [2]. In practice, they prepared groups of solutions containing different acids ant they evaluated them. Subsequently, they found that the most appropriate sample was the one with 25% of polyacrylic acid since it resisted better the hydrolytic disintegration. Nevertheless, this solution was not perfect as it hardened too fast [2]. To compensate for the lower acidity of the polymer compared to the phosphoric acid used in dental silicate cements, glass was modified to exhibit higher basicity [4]. Then, 15 years after the investigation by Wilson and co-workers, dental cements were introduced commercially but with inferior properties than those available today [4].

It should be stressed that glass-ionomer cements were developed as a response to the limitations of earlier dental silicate cements. Then, they became an important group of restorative materials. Their historical development shows that their success is based on combining chemical adhesion, fluoride release and acceptable biocompatibility in a single material. Because the behaviour of glass-ionomer cements is strongly dependent on their formulation, the composition of conventional GICs must be considered in detail. Today, GICs have multiple dental applications such as restorative materials, lining or base materials, fissure sealants and for orthodontic bracket bonding [1].

2.2. Composition, setting reaction, and maturation of conventional glass-ionomer cements

To define the composition of GICs not only must components be defined but also the form in which they are presented and the way in which they are put together. Firstly, the system consists of three components: polyacrylic acid, calcium fluoroaluminosilicate glass powder and water [4], [1]. These three ingredients can be presented in two different ways. On the one hand the entire acidic component may be incorporated into the glass powder, both in powder form and on the other hand, with part of the acid dissolved in the aqueous phase. Nevertheless, no significant differences in the final properties have been observed [1].

Glasses

The glasses used for GICs are alumino-silicate glasses, which contain Al, Si and Ca. Additionally, glasses for GICs must be basic and capable of reacting with an acid to create a salt. Their basicity characteristic is due to the fact that they are made of alumina and silica. Si^{4+} and Al^{3+} form a tetrahedral geometry with oxygen bridges. If they were only based on silica they would lose reactivity and basicity. By adding alumina, negative charges are created, which are compensated by Na^{+} and Ca^{2+} cations, consequently solving this problem and making the glass more basic [17], [1].

Phosphate and fluoride addition is needed too, more specifically fluoride, as this component is crucial to ensure glass reactivity and mechanical properties. Furthermore, when polyalkenoic acid attacks the glass during setting, F^{-} is released, creating a reservoir, that will release F^{-} in the future (this is of interest for dentistry) [17]. Initially, fluoride was added in flux form to decrease melting temperature. However, commercial glasses are based on calcium compounds with added sodium [17], [1].

Polymers

Going more into detail with polymers, polyalkenoic acids are the acids used for glass-ionomer cements. There are two types: homopolymer and copolymer of acrylic acid. In practice, no difference has been observed between these acids in terms of performance. Nevertheless, it is important to mention the difference regarding their internal changes during the first weeks after setting [1]. In the first place, homopolymers progressively improve their compressive strength during the first 4 to 6 weeks after setting. However, at the beginning copolymers undergo an increase in their strength, but then this value decreases down to a plateau. the higher crosslink density of copolymers [1].

Regarding polymers molecular mass, Wilson et al. discovered that polymer chain length is one of the most important parameters. It has been observed that a higher mass means a higher cement strength, but in contrast there is an increase in viscosity, which means that it is more difficult to mix components. The optimal weight has been proved to be between 11,000 and 52,000 [2] [1].

Additives

Tartaric acid has proven to be the most effective additive due to its ability to increase the working time (making the cement easier to mix), decrease the setting time, improve compressive strength and improve the resistance to acid dissolution. Tartaric acid avoids the creation of aluminium salts by chelating Al^{3+} ions. This delays the formation of ionic crosslinks involving Al^{3+} [1], [2].

Overall, the composition of conventional glass-ionomer cements explains many of their final properties. The glass phase, the polyacid, the water content and the additives do not act independently. They act as a system in which small compositional changes can modify many aspects such as reactivity, handling and mechanical performance. Therefore, the study of composition is not only descriptive, but also necessary to understand the later setting process and the final behaviour of the cement. Moreover, the powder (acid and glass) /liquid ratio and particle size are critical, as higher P/L ratio increases setting time and strength. Regarding particle size, smaller size means lower setting time but higher compressive strength [18], [17].

Setting reaction

When solid (glass) and liquid phase are brought into contact, a chain of setting reactions is activated. The components can be mixed manually using a spatula, or mechanically, by using capsules in which the components are separated by a membrane that is broken immediately before mixing [1].

The hardening process of GICs has been studied using different techniques, such as infrared spectroscopy, electron probe microanalysis and pH measurements [4]. These studies show that the hardening process is an acid-base reaction. When the acid and the glass are mixed, the acid attacks the glass and causes the release of ions such as Ca^{2+} and

Al^{3+} into the aqueous phase. These ions then help to form cross-linked polyacrylate structures, which are responsible for the setting of the material. This process usually takes place during the first few minutes after mixing [4], [19].

The protons of the acid attack the calcium-rich areas because calcium is located in areas where Al^{3+} has replaced Si^{4+}. First, Al^{3+} is more basic, and secondly, Al^{3+} needs more cations to reach energetic equilibrium. After this attack by the protons, Ca^{2+}, Na^{+}, and Al^{3+} are released. Studies including Fourier-transform infrared spectroscopy have shown that calcium is the one that is released into the matrix first, and then aluminium [4]. Once released, they react with the polyacid molecules to form ionic crosslinks, forming a solid structure. Regarding the aqueous phase, after this reaction it remains incorporated into the mixture in a homogeneous way, without there being distinction between the phases [1], [4].

The initial setting stage is a gelation process that occurs between the cement components. After that, the glass particles that do not react remain inside the silica (SiO_2) gel matrix and act as fillers. The material then hardens because the polymer chains of the polyacid become cross-linked with calcium and aluminium ions coming from the powder component [20]. Overall, glass-ionomer cements can be understood as materials made up of inorganic and organic networks, where the unreacted glass remains dispersed within the final structure [4].

To summarize, the reaction begins when the polyalkenoic acid interacts with the glass particles. This is followed by changes in the polymer chains and the release of metal ions into the aqueous phase. These ions contribute to the hardening of the cement, and chemical adhesion to the dental hard tissues is gradually developed (*Figure 1*) [3].

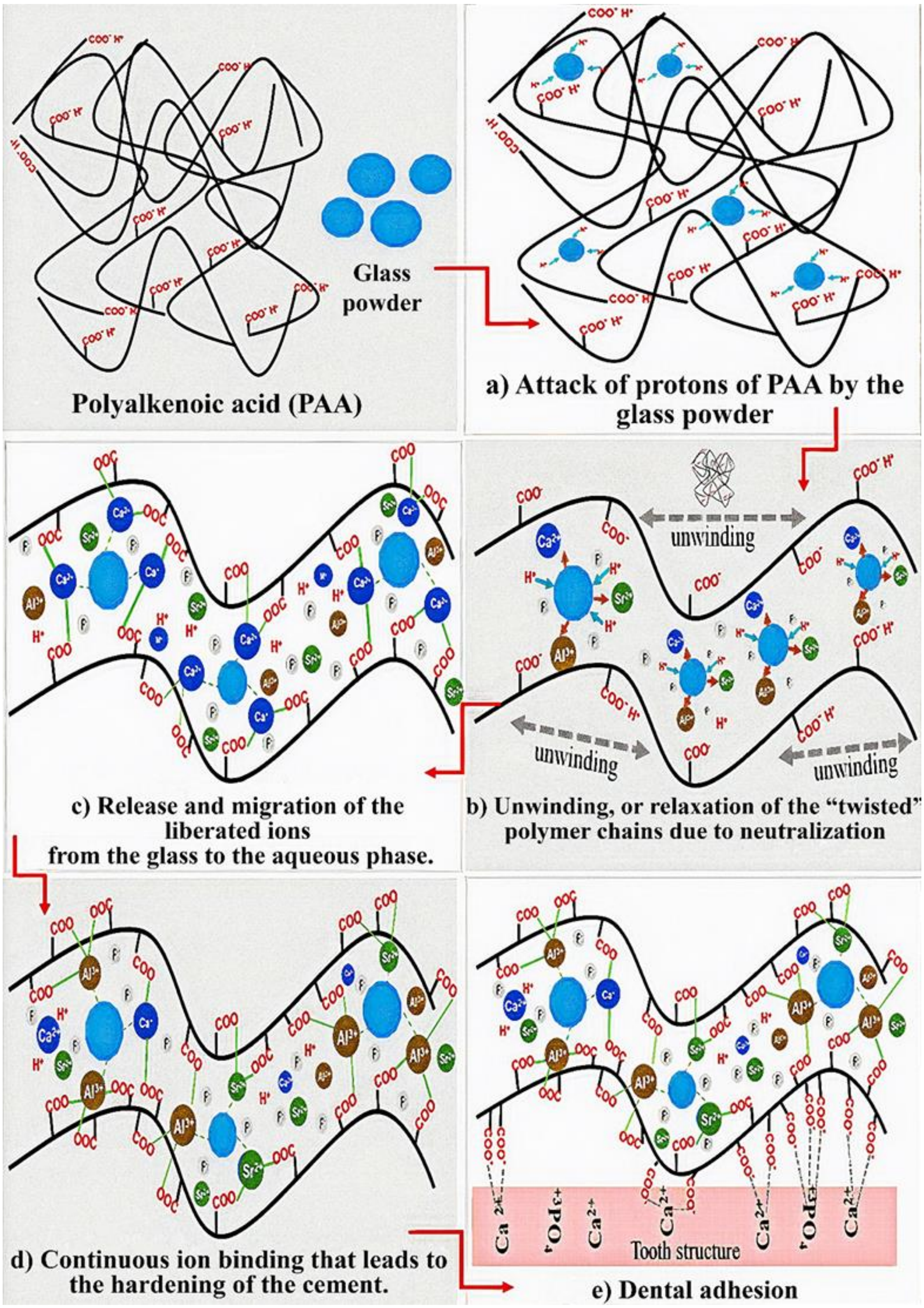


***Figure 1**: GICs acid-base reaction [3].*

Maturation

After the initial hardening, changes in the material continue to occur even months later. The compressive strength continues to increase until it reaches a stable value after a week (*Figure 2*), and the material becomes more brittle. Four months after initial hardening, glass ionomer cement can loose up to 75 MPa of compressive strength [5].

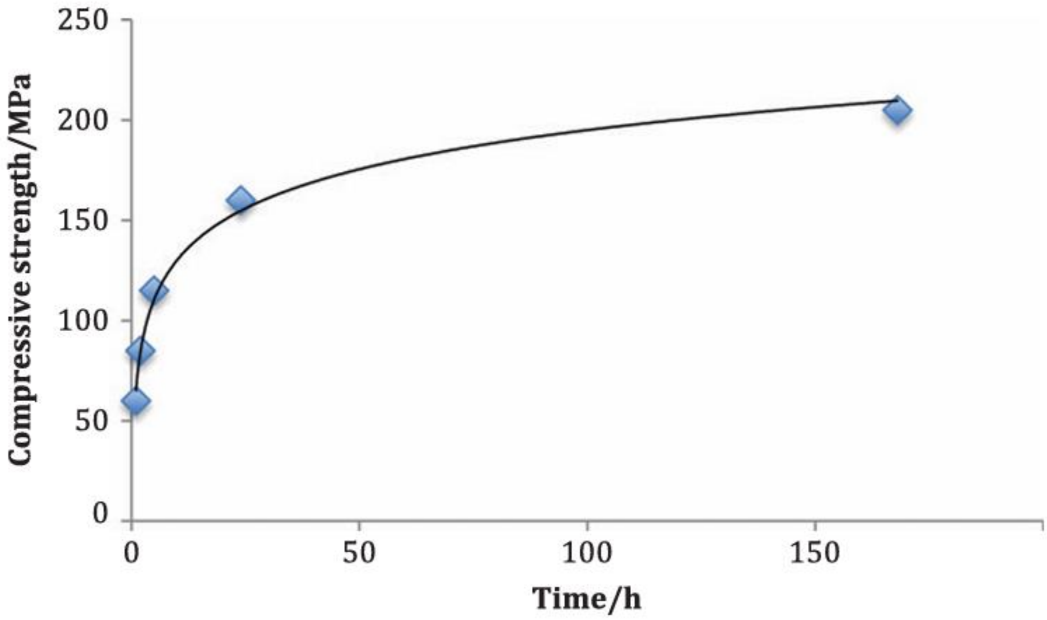


Figure 2: Evolution of compressive strength after initial setting [5].

Regarding the interaction with light, the cement loses opacity. There is also a progressive release of ions on the contact surface with the dentine. In addition, the proportion of bound water increases to some extent [5]. Nevertheless, these changes are not always the same because they depend on the chemical composition, as for example the particle size and shape [21]. Therefore, the initial setting stage should be considered only the beginning of the evolution of the material.

One important aspect is the change in the ratio of bound to unbound water during the months following the initial setting. Bound water is the water that remains in the structure even when the glass ionomer cement is exposed to dry and high-temperature conditions, including treatment with sulfuric acid. Unbound water, on the other hand, is the one that does disappear. With the aging of the cement, the amount of unbound water decreases due to the interaction with the acid and the glass [4], [20]. To summarize, water is important not only in setting, but also during posterior maturation.

Regarding final microstructure of matured cement, it is made of inorganic and organic networks that form a matrix with unreacted glass particles inside it [3], [4]. This unreacted glass serves as a reinforcing phase that makes the hardened cement stronger [1]. *Figure 3* shows the structure of a typical glass-ionomer cement. Label 1 shows the glass particles that did not react during the setting process. Label 2 indicates the siliceous hydrogel layer located between those particles and the surrounding material. Label 3 represents the polysalt hydrogel matrix around them. It can also be seen that the glass particles contain different amounts of radiopaque nanoparticles [22]. This microstructure explains why the unreacted glass works as a reinforcement in the hardened cement. The particles that do not take part in the reaction help strengthen the material by limiting the advance of cracks through the cement [23].

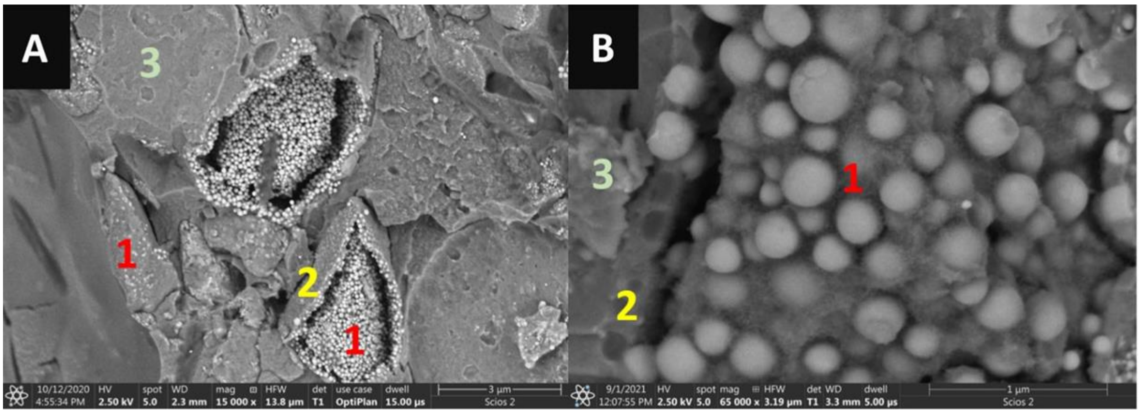


***Figure 3**: GICs BSE images taken at magnifications of 15,000× in image A and 65,000× in image B [22].*

Before finishing this section, it is worth mentioning the secondary setting reaction. There is another type of cement known as pseudo cements, which are not formed with polymeric acid but with others such as acetic and lactic acids. These pseudo cements, after being hardened for several hours, show an important characteristic, which is insolubility. Studies have shown that this is due to the formation of an inorganic network in ion-deficient areas after the crystal reaction. Since this inorganic network only forms with crystals that contain phosphates, it is concluded that this network is a type of phosphate species. When this type of cement is compared with those formed from polymeric acid, it is observed that initially, the formation of this secondary inorganic network reinforces the cement and makes it chemically more stable. But the same network, when it continues developing or maturing over time, can make the structure more rigid and less plastic, and that is why the material ages by becoming more brittle [4], [5].

2.3. Functional properties of glass-ionomer cements: adhesion, bioactivity, fluoride release and antibacterial behaviour

Adhesion and bioactivity

Glass ionomer cements (GICs) are widely used in dentistry due to their biocompatibility and chemical bonding to dental tissues, because the reaction between fluoroaluminosilicate glass and polyalkenoic acid creates a material capable of adhering to enamel and dentin. They also have the ability to release fluoride ions which are known for their anticaries properties [3].

Going more into detail about the bond between the cement and the tooth, it occurs in two stages. In the first stage, a weak union of hydrogen bridges forms between the carboxylate groups of the cement and a very tightly bound superficial layer of water on the tooth. In the second stage, this weak bond changes to a strong bond between the carboxylate groups and the calcium ions present in the apatite crystals on the surface of the tooth, thus forming an ion exchange surface (*Figure 4*). The existence of this surface has been confirmed by infra-red spectroscopy (*Figure 5*) [20], [24]. Therefore, adhesion not only depends on the initial union, but an interface must also be developed progressively.

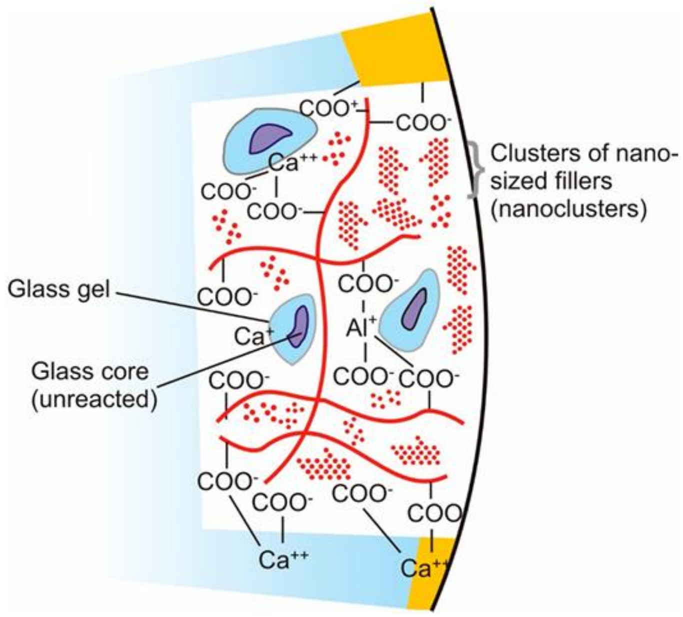


Figure 4*: Adhesion mechanism between contact surfaces of the cement and the tooth [20].*

Figure 5: *Ion exchange surface seen by infrared spectroscopy [24].*

This interface, which develops over time, is a consequence of the bioactivity of GICs [1]. Another product of this property is the antibacterial effect of these cements [20]. In a comparative study between GICs and composites, it was observed that GICs reduce the appearance of the bacterium *Streptococcus mutans [25]*. This is due to the presence of aluminium fluorosilicate glasses, which lead to the long-term release of fluoride, an element with an anti-cariostatic effect [26], [20]. This property makes glass-ionomer cements very useful for endodontics and restorative dentistry, since caries are known to be a global problem among population [27], [16].

To determine the potential deterioration of the bond between glass ionomer cement and dentin, a study was conducted by Hoshika et al. (2021). In this study, samples of glass ionomer cement bonded to dentin were created and stored for one year (*Figure 6*). After this period, they were analysed, and it was found that there had been no significant deterioration of the bond, maintaining the initial bond strength [28].

To sum up, the clinical value of glass-ionomer cements is not limited to simple adhesion, since their adhesion to dental tissues is linked to bioactivity and to the formation of a stable interface over time. This makes them especially useful in restorative dentistry, where both adhesion and biological compatibility are important. In this sense, the ability of GICs to form a durable bond with dentin is one of the main reasons which explains why these materials continue to be widely studied and used.

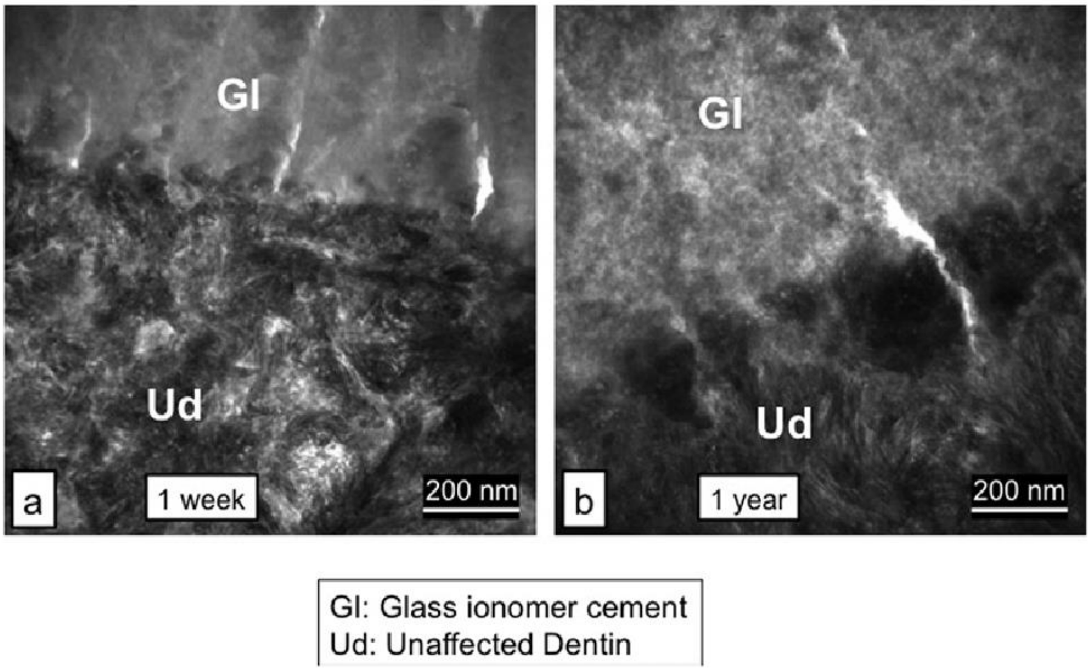


Figure 6: Bond between GIC and dentine after 1 year [28].

Fluoride release and antibacterial behaviour

Fluoride release is one of the most important characteristics of glass ionomer cements. They act as a fluoride source that helps maintain higher fluoride concentrations in saliva, dental plaque and tooth [11], [29]. However, fluoride release is not a gradual process. After the initial setting, there is what is known as the initial "burst," followed by a reduction in the release rate. This does not mean that fluoride release stops, but rather that it continues to do so, albeit at a lower rate [29]. Therefore, their use in dental implants helps prevent secondary caries [11], [29].

In a study conducted by Mousavinasab and Meyers, four conventional glass ionomer cements (GICs) were compared with a compomer known as Dyract Extra and a giomer called Beautifil (compomers and giomers are resin-based restorative materials). The results showed that the conventional glass ionomer cements had a significantly higher fluoride release than the compomer and giomer [29]. However, this protection is not always sufficient, which is why various ways to improve them have been researched since their creation [11]. A study conducted by Łuczaj-Cepowicz et al. demonstrated that the antibacterial behavior of glass ionomer cements (GICs) is not identical across all GICs [30]. Two Fuji and two Ketec glass ionomer cements were compared against Streptococcus and Lactobacillus bacteria. The study showed that while the GICs performed well against some bacteria, there were others against which they were ineffective. This was the case with Ketec GICs against the bacterium L. casei [30].

To improve these antimicrobial properties, various additives have been sought for GICs. However, not all of them are effective, as in some cases they improve effectiveness against bacteria at the expense of worsening mechanical properties [11].

The sum up, fluoride release is one of the main functional advantages of glass-ionomer cements, since it gives them an anticariogenic and partly antibacterial role. As explained before, other restorative materials do not always show these properties to the same extent. Nevertheless, this protective effect is not identical in all glass-ionomer cements and is not always enough on its own. For that reason, fluoride release should be understood as an important benefit of GICs, but not as a complete solution to their clinical limitations. Therefore, further studies to improve GICs properties should be done.

2.4. Mechanical properties, ageing behaviour, and limitations of conventional glass-ionomer cements

Despite all these upsides regarding biocompatibility mentioned above, traditional glass ionomer cements have deficiencies in terms of mechanical properties, and they do not respond well to high stress requirements [6] [7]. Poor resistance to fracture, low toughness, limited wear resistance and high sensitivity to moisture are some of these drawbacks [6], [8]. Studying the compressive strength of GICs is of crucial importance to understand their mechanical integrity [6]. Therefore, this is one of the main focuses of this project.

In a comparative study that was carried out between different glass ionomer cements (Fuji IX, ChemFil Rock Riva Self-Cure and Ketac Nano), it was seen that their compressive strength is approximately in the range of 90 to 170 MPa, with ChemFil Rock being the most resistant of them [6]. However, even the largest value in this range is below the actual strength of human dentin, which depending on the study carried out has values from 190 MPa [31] to 585 MPa [32]. It should be noted that compressive strength values vary greatly depending on the study, which is why the range is so wide. Regarding fracture resistance, conventional GICs typically perform worse than natural teeth (Table 1). In some studies, GIC fracture resistance is around 0.30 MPa $m^{1/2}$, while enamel reaches 3.93 MPa $m^{1/2}$ and dentin 2.3 MPa $m^{1/2}$ [9]. Moreover, conventional GICs exhibit worse mechanical properties than the natural human tooth (*Table 1*). Regarding their thermal expansion, GICs have a low coefficient, which is very similar to that of natural teeth [7]. This makes them very useful in clinical use.

__Table 1__: Mechanical properties of dentine, enamel and conventional GICs

Material	Property	Value	Reference
Dentine	Compressive strength	190 - 585 MPa	[31], [32]
	Fracture toughness	1.1 - 2.3 MPa $m^{1/2}$	[9]
Enamel	Compressive strength	62.2 - 363 MPa	[31], [32]
	Fracture toughness	0.67 - 3.93 MPa $m^{1/2}$	[9]
Conventional GICs	Compressive strength	90 - 170 MPa	[6]
	Fracture toughness	0.18 - 0.30 MPa $m^{1/2}$	[9]

Microstructure governs a key role because mechanical properties are strongly influenced by factors such as the quality of the bond between the glass particles and the polymer matrix, as well as by particle size and the amount and dimensions of voids [7]. The smaller the particle size, the greater compressive performance, wear resistance and surface hardness of the material [17], [18] . Another factor that is directly related to the mechanical resistance of cement is the powder/liquid ratio, because the greater the amount of powder, the greater the resistance of the material [17]. Also, it should be noted that the mechanical behavior does not depend only on the formulation, but also on the storage environment during the maturation stage [22].

In summary, conventional glass-ionomer cements have several useful properties. Nevertheless, their mechanical weakness still limits their use in demanding situations. Their performance depends not only on strength values, but also on factors such as fracture resistance, particle size, powder-to-liquid ratio and storage conditions. Therefore, these limitations justify the search for modified GIC formulations with improved mechanical behaviour.

2.5. Strategies for glass-ionomer cements

As explained in the previous section, the shortcomings in mechanical performance of conventional GICs restrict their applications in restorations exposed to low stress. Therefore, various methods have been studied to improve them in this aspect, including the addition of fibres, nanoparticles, larger filler particles, water-hardening systems, resin-modified glass-ionomer cements (RMGICs) (which include both self-curing and light-curing systems), and amino acid residue-modified glass ionomers [7], [8], [9].

Metal reinforcement

The first attempts to improve the mechanical properties of GICs by adding metal reinforcements were made with chromium, aluminium, nickel-aluminium alloy, and silver-tin alloy. Of these, silver-tin alloy was reported to offer the greatest improvements, increasing flexural strength by up to 30 MPa. Because of this, GIC enhanced with this alloy was marketed and became very popular, becoming known as "*Miracle Mix*". Subsequent studies concluded that its mechanical improvement over the original cement was unclear, however, it continues to be used because it is radiopaque [9].

Reinforcement with fibres

Attempts have also been made to improve the performance of GICs by adding carbon, alumina and reactive glass fibres. Regarding carbon and alumina fibres, both demonstrated improvements in flexural strength, with carbon fibre showing a slight advantage (increasing from 10 MPa to 53 MPa). Furthermore, carbon fibre does not increase brittleness, unlike alumina fibre. In relation to reactive glass fibres, they also improve flexural strength about 6 MPa, by increasing the work of flow [9].

Bioactive and nanoparticulate modifications of the powder phase

Nano-hydroxyapatite or nano-fluorapatite have also been used as additives to conventional GIC powder due to their affinity for human bone and their ability to highly improve mechanical properties of material (compressive strength, tensile strength, and flexural strength). Of the two, nano-fluorapatite has been shown to exhibit superior adhesion. The improvement in mechanical properties is due to the chemical reaction between polyacrylic acid and apatite crystals. The good adhesion of GICs containing these additives is attributed to the large contact surface area of the nano-apatite particles, resulting from their minute particle size [8].

Titanium dioxide (TiO_2) and zirconia nanoparticles have also been used to improve the mechanical properties of GICs. Regarding TiO_2, although it is toxic, in practice it has not been shown to increase the toxicity of GIC. As for zirconia, adding too much increases brittleness [8].

Apart from nanoparticles, the addition of inorganic powders with bigger particle sizes such as bioactive glass (BAG) has been shown to improve the bioactive properties of cement [9], [12]. However, the greater the amount of bioactive glass added, the worse the mechanical performance. The addition of Al^{3+} ions to BAG improves mechanical performance but worsens bioactivity [12]. In conclusion, the addition of these inorganic powders, despite improving bioactivity, does not significantly improve mechanical properties, even reducing compressive strength in some cases [9].

Antimicrobial additives

One modification approach has been to introduce antimicrobial additives such as chlorhexidine diacetate or chlorhexidine digluconate. Both have been shown to have antimicrobial effects when added to glass ionomers. These additives disperse within the GIC by diffusion, but a large portion remains trapped inside the cement. They typically increase setting time but worsen mechanical properties. However, in small quantities, they do not significantly affect the properties [11].

The use of silver nanoparticles as an additive has also been studied, demonstrating a great improvement in terms of antibacterial properties and also improving mechanical properties, making this a more promising additive [9].

Special formulations and processing-assisted modifications

If, instead of using regular glass, glass that has been pre-treated with strong acids such as hydrochloric acid is used, and which also contains silicon oil and nano-hydroxyapatite and nano-fluorapatite fillers, the result is known as "Glass Carbomer." This glass is much more bioactive. Regarding its mechanical properties, it becomes more brittle, although the silicon oil mitigates this effect [8]. Glass carbomer as a distinct formulation strategy, not simply another filler addition.

Another modification that can be performed on glass-ionomer cements is the external transmission of energy to their surface, such as radiant heat through light-curing or kinetic energy through ultrasound. This technique improves the setting reactions, allowing mechanical properties of cement to be achieved more quickly, which, as mentioned in previous sections, take days or even months to attain [8]. They therefore improve surface hardness, compressive strength, adhesion, and abrasion resistance. Conversely, they worsen fluoride release and solubility [8].

Resin-Modified Glass Ionomer Cements

Resin-Modified Glass Ionomer Cements (RMGICs) are a modification of conventional GICs that emerged in the late 1980s. They were created to improve the mechanical properties and moisture sensitivity of GICs. Their chemical composition differs from that of conventional cements in that they contain certain added monomers, and they also require an external element for polymerization [8], [33].

Their formation involves two phases. The first is the acid-base reaction that occurs after mixing the solid and liquid phases. The second is polymerization, which must be induced by a light-curing process[8].

Regarding their advantages over conventional GICs, RMGICs possess superior mechanical properties, such as higher compressive strength, better moisture resistance, and improved optical properties, as their greater translucency results in a better finish. However, they also have some drawbacks, including decreased fluoride release and shrinkage during solidification. This last point means that they decrease in volume, which can affect the final result [8].

Concluding remarks

In conclusion, since the creation of glass ionomer cements, many modification methods have emerged to improve their mechanical properties (*Table 2*). However, few have demonstrated a significant improvement. Those that have shown the best results are those involving added fibers and those incorporating certain nanoparticles. It is important to note that each researcher uses different geometries and conditions for their samples, making it difficult to reach a consensus within the scientific community. Therefore, International Standards should be followed [9].

Table 2: Main modification strategies of GICs and their limitations

Modification strategy	Representative examples	Advantages	Main limitations	Ref.
Metal reinforcement	Chromium, aluminium, nickel-aluminium alloy, and silver-tin alloy	Improved radiopacity; some studies reported moderate improvement in strength	Mechanical improvement is inconsistent	[9]
Fibre reinforcement	Carbon, alumina and reactive glass fibres	Improvements in flexural strength	Some fibres may increase brittleness	[9]
Bioactive and nanoparticulate powder-phase modification	Nano-hydroxyapatite, nano-fluorapatite	Highly improved mechanical properties and better adhesion		[8]
Other nanoparticle additions	TiO_2, zirconia	Potential improvement in mechanical properties	Excessive TiO_2 is toxic and too much zirconia increase brittleness	[8]
Bioactive glass additions	BAG	Improved bioactivity	Too much BAG damages mechanical properties	[9], [12]
Antimicrobial additives	Chlorhexidine diacetate and chlorhexidine digluconate	Antimicrobial effect	Worsen mechanical properties	[11]
Resin-modified glass-ionomer cements (RMGICs)	Light-curing RMGICs	Better mechanical properties, better moisture resistance, and improved optical properties	Decreased fluoride release and shrinkage	[8]
Glass carbomer	Pre-treated glass with silicon oil and nano-hydroxyapatite	Better bioactivity	More brittle cement	[8]
Processing-assisted modification	Radiant heat, light-curing, ultrasound	Mechanical properties achieved more quickly	Worse fluoride release and solubility	[8]
Metallic antimicrobial nanoparticles	Silver nanoparticles	Antibacterial properties	Effect depends on dose and formulation	[9]

2.6. Effect of aging and hydrothermal conditions on the functional properties of glass-ionomer cements

Not only is the composition of the glass ionomer cement important, but also the conditions it experiences during its curing process. After application in the oral cavity, the material is subjected to various temperature changes (up to 50 cycles per day), which generates stresses within the polymer matrix on the surface of material [22], [34]. Consequently, its mechanical properties are affected. Studies have shown that these properties change compared to those of glass ionomer cements cured under constant temperature conditions [22].

To measure properties such as microhardness, compressive strength, wear behaviour, and surface roughness, a study conducted by Łępicka et al. investigated two groups of samples composed of different GICs (Riva Self Cure (SC), Ketac Universal (KU), Ketac Molar Easymix (KME)), one of these groups were aged at a constant temperature of 37°C in deionized water, and the other underwent 20,000 cycles with temperature changes between 5°C and 55°C [22]. Hydrothermal fatigue changed the functional properties of conventional GICs, but these changes were not always negative and depended on the cement formulation. Regarding the properties negatively affected by hydrothermal aging, for KU, these were its microhardness, compressive strength, and roughness. For RSC, its microhardness and roughness were also negatively affected. Conversely, the properties positively affected by hydrothermal aging in KME were its microhardness, compressive strength, and roughness [22].

In conclusion, unlike studies that measure the mechanical properties of glass-ionomer cements matured under constant conditions, those done on cements exposed to cyclical temperature changes do not show an increase in strength over time [22].

2.7. Lanthanide-based modifications in biomaterials

Currently, seventeen rare earth (RE) elements are known. Rare earth elements are found in granite, basalt, and silicate ores, in very specific layers of the Earth crust. Lanthanides are among the most well-known rare earth elements. These materials are known for their unique spectroscopic properties [35].

One of these properties is photoluminescence, caused by the special behaviour of their ions in terms of luminescence and electromagnetism. RE elements are capable of both converting high-energy light into low-energy light (downconversion) and vice versa (upconversion). This property makes them very useful in biomedical applications such as diagnostic procedures and radiotherapy because their light stands out better, penetrates tissues more effectively, and produces less interference, making them easier to detect [35], [36].

In nanoscience, rare earth elements can be used as the primary element for manufacturing nanomaterials or as dopants. As mentioned previously, these nanomaterials composed of rare earth elements are widely used in diagnostics and imaging. Furthermore, lanthanide nanoparticles show great promise in image-assisted surgery [36]. The two main *in vivo* applications of these materials in diagnostics are planar and tomographic fluorescence and bioluminescence imaging. However, they are mainly limited to pre-clinical applications due to factors such as their toxicity [36]. Regarding tissue repair, rare earth metals have shown promise in repairing both bone and soft tissue, even improving fracture resistance. Furthermore, when used as nanoparticles, they enhance biocompatibility [37].

Getting to the heart of this section, one of the most promising clinical practices is doping hydroxyapatite crystals with lanthanides, since hydroxyapatite is the inorganic component found in the greatest amount in bone tissue. The drawbacks of hydroxyapatite are its deficiencies in terms of mechanical properties and its low antimicrobial properties. It is known that one of the most frequent causes of implant failures is the formation of bacteria on the surface due to the poor antimicrobial capacity of hydroxyapatite nanoparticles. Therefore, incorporating lanthanides as a doping agent can improve these deficiencies. Lanthanides are able to replace the sites occupied by calcium, resulting in apatite having lower crystallability, smaller crystal size, and greater solubility. Consequently, Lanthanide substitution can improve the hardness of calcium phosphate and also provide photoluminescent properties and higher electrical conductivity [38].

Table 3: Changed produced in Hydroxyapatite by lanthanides [38].

Lanthanides	Main effects
Samarium	Reduced crystallinity, photoluminescence
Lanthanum	Higher crystallinity / hardness
Cerium	Photoluminescence / crystallinity changes
Gadolinium	Electrical conductivity, photoluminescence
Erbium	Bioimaging-related optical behaviour

Regarding biomedical uses of lanthanide-substituted hydroxyapatite crystals, lanthanide-doped HAp can improve bone regeneration, reduce inflammation and oxidative stress, and add antimicrobial activity. This material has uses such as drug delivery systems, creation of bone repair materials, and theragnostic platforms. However, it is also necessary to consider the maximum percentage of this element that can be used to avoid toxicity to humans [38].

The following table (*Table 4*) shows the main biomedical uses of lanthanide-substituted hydroxyapatite crystals and the main advantages they bring to these fields.

Table 4: Biomedical uses of lanthanide-substituted hydroxyapatite crystals [37], [38].

Lanthanides	Main biomedical uses	General properties and advantages
Samarium	Implant coatings, bone materials, theranostics, biomarker detection	Improve material performance and provide antibacterial effects
Lanthanum	Implant coatings, bone repair, wound healing, drug delivery, bioimaging	Support cell growth, tissue repair, and healing
Cerium	Implant coatings, bone regeneration, wound healing	Shows good biocompatibility and help bone and tissue repair
Gadolinium	Implant coatings, bone substitutes, bone/cartilage repair, theranostics	Promote cell response and tissue regeneration
Erbium	Bioimaging, drug delivery, radiosynovectomy	Mainly useful for imaging and some therapeutic applications

In relation to toxicity of rare earth elements, this is an issue that must be monitored. Depending on the amount of RE element used, there may be a benefit or harm to health [37]. The FDA has warned that the use of gadolinium in contrast agents can be dangerous because it can remain in the human body for months after the test. Furthermore, other lanthanides can replace the sites occupied by calcium in the intestine, thus causing problems. Lanthanum, in particular, can cause DNA damage. To address this, methods such as encapsulating lanthanides in nanoparticles that the body can quickly excrete through urine or feces have been proposed [36]. Finally, there are still not enough studies on the safety of using these elements in biomaterials, so further research is needed [37].

2.8. Samarium as a modifying ion in glass-ionomer cements: rationale and research gap

Among the lanthanides, samarium is of particular interest due to its greater economic accessibility, its special radioactive properties (fluorescence and high energy levels) and because it is one of the most abundant lanthanides [10], [13]. Furthermore, its antimicrobial properties and its affinity for bone tissue in the form of Sm^{3+} cations are well known. In this state, samarium is quite similar to calcium in terms of ionic radius (*Figure 7*) and the way it interacts with surrounding atoms. Therefore, it has been proposed as a doping agent in bioactive glass, giving it optical and mechanical improvements [10], [13], [14].

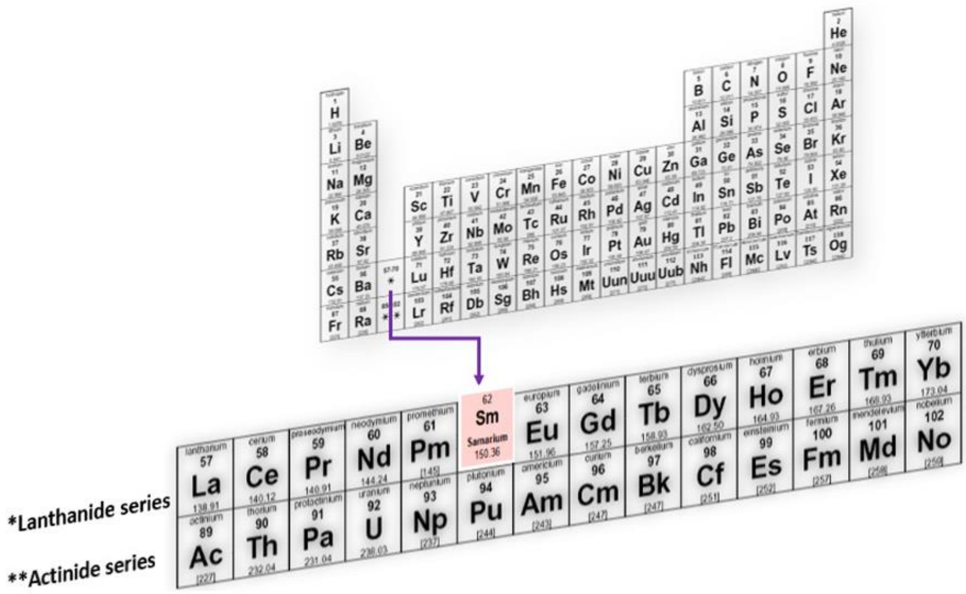


***Figure** 7: Location of samarium in the periodic table [10].*

Although it is still a material that requires further study, very promising results have been obtained when investigating its effect on the properties of the materials to which it has been added. Composites containing samarium-doped P_2O_5 glass-reinforced HAp have shown improved osteoblastic activity and good antibacterial performance. Moreover, the incorporation of samarium into zinc oxide nanoparticles (ZnO) was found to significantly improve their anticancer effect [13].

Regarding the addition of samarium to bioactive glasses, it has been shown to improve both their bioactive and mechanical properties. It has been observed to enhance their thermal stability, durability, and antibacterial properties. Consequently, this material is very useful for applications such as tissue engineering and bone repair [10].

When comparing hydroxyapatite (HAp) with samarium-doped hydroxyapatite (SmHAp), clear advantages are observed in the use of samarium. A study conducted using techniques such as X-ray diffraction demonstrated that the addition of samarium to hydroxyapatite improves both biocompatibility and cytocompatibility. This comparative study showed that the amount of HAp that can be used before a decrease in biocompatibility is 25 μg/mL, while for SmHAp it is twice that amount. Regarding cytocompatibility, this property is affected starting at 100 μg/mL for HAp, while for SmHAp it is four times higher. Furthermore, samarium is able to reduce the negative effect that hydroxyapatite has on calcium stability [13].

Another study analysing samarium-doped hydroxyapatite nanoparticle coatings (5-10 mol%) demonstrated high biocompatibility between these compound and human gingival cells, withstanding harsh conditions produced by external agents such as human saliva, acidogenic bacteria, fermentable carbohydrates [15], [39]. This is crucial because it directly relates to the focus of this research. These aggressive conditions to which the bone tissue of the oral cavity is exposed are the cause of its structural damage. Samarium-doped hydroxyapatite (Sm-HAp) has proven to aid in dentin remineralization, in addition to having over 85% biocompatibility [39].

Moreover, a study was conducted by Saud et al. in which samarium oxide (Sm_2O_3) was used as a component of ceramic used in dentistry demonstrated antimicrobial activity, radiopacity and acceptable biocompatibility [14]. Regarding the dangers of samarium, studies have shown its low toxicity, in addition to not affecting the behaviour of osteoblasts, cells responsible for growth, fracture repair and bone renewal [15].

In conclusion, although samarium appears to be a material with a promising future in biomedical applications, its precise mechanisms of action are not yet fully understood [10], [15]. Furthermore, samarium-containing bioactive glasses still require improvement in certain aspects, as samarium ions can affect the structure and thermal properties of the glass, unintentionally altering other properties [10].

Table 5: Representative samarium-containing biomaterials and their reported effects relevant to dentistry and bone-related applications.

Application	Main reported effects	Relevance	Ref.
Sm-doped hydroxyapatite nanoparticles	Maintained the apatite structure and showed good biocompatibility. Antibacterial activity and positive, although limited, effect on osteogenesis and cell migration.	Samarium may improve the biological behaviour of calcium-based biomaterials and could also contribute to their antimicrobial activity.	[10], [13], [14]
Sm-HAp coatings	Antibiofilm activity and good biocompatibility. No clear cytotoxic effects on human gingival fibroblasts.	Relevant in dentistry. Possible applications related to implants.	[15], [39]
Sm-HAp in dentin remineralization	High cell viability and helped promote dentin remineralization.	Dental relevance, because it suggests that samarium-containing biomaterials may be useful for mineralized dental tissues.	[39]
Sm-doped bioactive glasses	Improved bioactive, optical and mechanical properties	Use of samarium is not restricted to hydroxyapatite-based systems. Improve glass-based biomaterials, which are more closely related to the glass phase studied in this work.	[13]
Sm_2O_3-containing endodontic cement	Acceptable biocompatibility, antibacterial activity, radiopacity, and the ability to form an apatite layer.	Especially relevant because it shows that samarium oxide has already been studied in cement-like materials for dental use.	[14]

Nowadays, among all the existing modifications to improve the strength of glass ionomer cements, not all work properly. The focus has shifted to nanoscale particles and BAGs (bioactive glass-reinforced glass ionomer cements) [9], [40].

As explained before, samarium is a plausible candidate modifier for GICs because it has already shown promising behaviour in hydroxyapatite, coatings, bioactive glasses and cement-like dental systems.

The evidence related to samarium is mostly found in studies about hydroxyapatite, coatings, bioactive glasses and endodontic materials, rather than in conventional glass-ionomer cements. In these fields it has shown promising behaviour. For this reason, the present study addresses this gap by evaluating glasses in which SiO_2 is partially substituted by Sm_2O_3 and then used as the powder phase for experimental GICs.

3. Objectives

The main objective of this study is to evaluate the effect of samarium oxide (Sm_2O_3) incorporation on experimental glass-ionomer cements, specially in their mechanical and structural properties. To achieve this, four groups of specimens were manufactured at Białystok University of Technology. These groups are based on the SiO_2 - Al_2O_3 - P_2O_5 - CaO - CaF_2 glass system, differing only in their samarium content:

- Reference (Sm0): no samarium oxide
- Sm0.5: 0.5 mol% SiO_2 substituted by Sm_2O_3
- Sm1: 1 mol% SiO_2 substituted by Sm_2O_3
- Sm5: 5 mol% SiO_2 substituted by Sm_2O_3

The objectives pursued in this work are the following:

First of all, to determine the compressive strength of each experimental group in accordance with ISO 9917-1. An universal testing machine will be used for this purpose. This test is performed in order to assess whether samarium content influences the compressive strength capacity of the cement.

Second, to characterise the surface microhardness of all groups using the Vickers hardness test. This process is done to evaluate the effect of Sm_2O_3 concentration on surface mechanical resistance.

Third, to measure the surface roughness of the specimens using a confocal laser scanning microscope. In this way, it can be known whether samarium incorporation modifies the surface topography of the hardened cement.

Fourth, to analyse the fracture surfaces and internal microstructure of the specimens using scanning electron microscopy (SEM). For this purpose, the broken specimens from the compressive strength test will be used in order to identify microstructural differences between the reference and samarium-doped groups.

Finally, an additional step motivated by the compressive strength results obtained during the main experimental phase will be carried out. It will be explored whether hydrochloric acid pretreatment of the glass powder could improve the mechanical behaviour of the resulting cement.

Based on all these results, the influence of samarium modification on the functional properties of the experimental cements was discussed and interpreted in the context of the existing literature on glass-ionomer cements and lanthanide-containing biomaterials.

4. Materials and Methods

4.1. Materials

The glass powder used for the preparation of the glass ionomer cement (GIC) was synthesized at Białystok University of Technology. The base composition was designed according to the fluoroaluminosilicate glass system reported by Abo-Mosallam and Park [41]. Four glass compositions based on the SiO_2-Al_2O_3-P_2O_5-CaO-CaF_2 system were prepared, containing 0, 0.5, 1, and 5 mol% Sm_2O_3, respectively (Table 6). Samarium oxide was introduced by partially replacing SiO_2 on a molar basis. The aim of this modification was to evaluate the effect of samarium oxide addition on the mechanical properties of the resulting glass ionomer cements.

The glasses were produced using a conventional high-temperature melting method. Homogenized batches of high-purity powder reagents, with a purity of at least 99.99%, were placed in a platinum crucible and heated in an electric resistance furnace at a rate of 10 °C/min up to 1500 °C. After reaching the target temperature, the melt was held for 3 h to ensure complete homogenization. The molten glass was then rapidly quenched in cold water.

The obtained glass was ground in a Fritsch Pulverisette 7 premium line planetary ball mill using a zirconia grinding bowl and zirconia balls. Dry milling was carried out for 20 min at 500 rpm with a ball-to-powder ratio of 10:1. The milled powder was then dry-sieved using laboratory sieves to obtain a reproducible particle size distribution suitable for GIC preparation.

The liquid component used for cement preparation was the acid solution from Ketac Universal Glass Ionomer Restorative Material Radiopaque, marketed by 3M ESPE. Four experimental groups were prepared, corresponding to the different Sm_2O_3 contents in the glass powder, as shown in Table 6. The reference group contained no samarium oxide, whereas the modified groups contained 0.5, 1, and 5 mol% Sm_2O_3.

The same powder/liquid ratio was used for all experimental groups to ensure that the observed differences in properties were related to the Sm_2O_3 content rather than to variations in cement consistency. A powder/liquid ratio of 1.34 was applied, corresponding to one scoop of powder per one drop of liquid. The components were mixed manually according to the same procedure for all samples.

The GIC used for this research was manufactured at the Białystok University of Technology. The liquid acid solution was the Ketac Universal Glass Ionomer Restorative Material Radiopaque, marketed by 3M ESPE. Four experimental groups were prepared (*Table 6*), each with a different amount of samarium oxide (Sm_2O_3). The first group, serving as the reference, contained 0 mol% of samarium oxide. The remaining groups contained 0.5 mol%, 1 mol%, and 5 mol%.

Regarding the powder/liquid ratio, the same ratio was used in all groups so that the results would depend on the amount of samarium and not on the powder/liquid ratio, which is directly related to the mechanical properties. The ratio was 1,34 and consists of one scoop of powder per one drop of liquid, subsequently mixed by hand.

***Table 6**: Experimental groups prepared in this study*

Group	**Samarium content (SiO_2 substitution, mol%)**
Sm0 (reference)	0
Sm0.5	0.5
Sm1	1
Sm5	5

4.2. Sample preparation

The instructions provided by the liquid acid solution manufacturer were followed to prepare the mixture of glass powder and aqueous acid solution. At a room temperature of 20-25°C, a metal spatula was used to manually mix the mixture for approximately 45 seconds until a homogeneous paste was obtained. Immediately after mixing, the paste was placed in the methacrylate mould. The mould was tightened with nuts to maintain the correct specimen geometry during the initial setting stage.

After filling the mould with the mixture, a plastic film was placed on top, followed by a glass plate. Pressure was applied to this glass plate with a weight to seal the mixture, also removing excess material and obtaining a flatter and more parallel specimen surface. The samples were then left to harden in the mould for 45 minutes, allowing the initial setting reaction to take place.

Following this process, the specimens were removed from the mould and placed in a container of deionized water. The four containers (each with its corresponding group of samples) were stored in an oven at 37°C for 7 days. This temperature was selected to simulate the maturation conditions they would encounter in the human oral cavity. After the maturation period, the specimens were polished before testing.

The prepared samples were divided into two groups, one for the compression test and another for the microhardness and roughness test. Each group had different geometry.

The samples for compression test had a cylindrical geometry. Their preparation took into account the main requirements of the compressive strength test described in ISO 9917-1 for water-based acid-base dental cements. For this purpose, a mould with several cavities to be filled was used, each measuring 6 mm in height and 4 mm in diameter. This geometry was also used by Łępicka et al., where cylindrical specimens of 4 mm in diameter and 6 mm in height were prepared for compressive strength testing according to ISO 9917-1 [22].

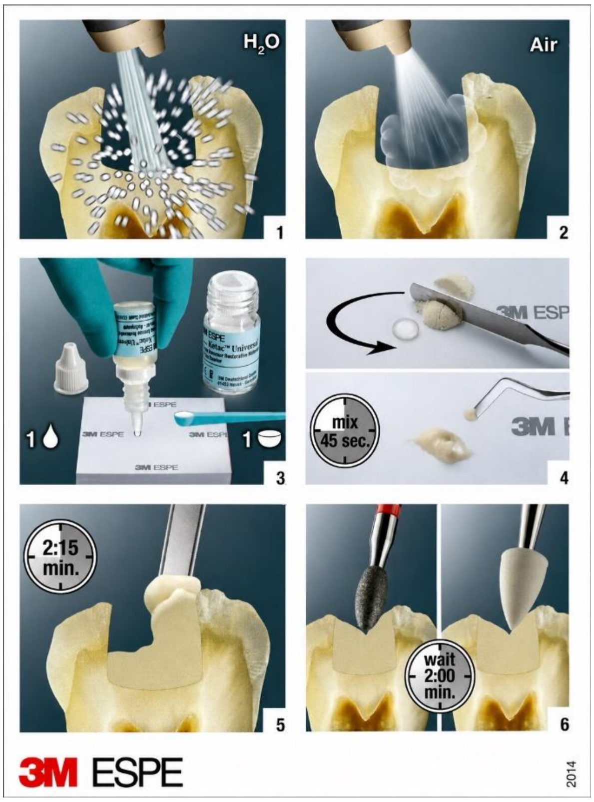


Figure 8: Sample preparation of the glass-ionomer cement specimens by hand-mixing

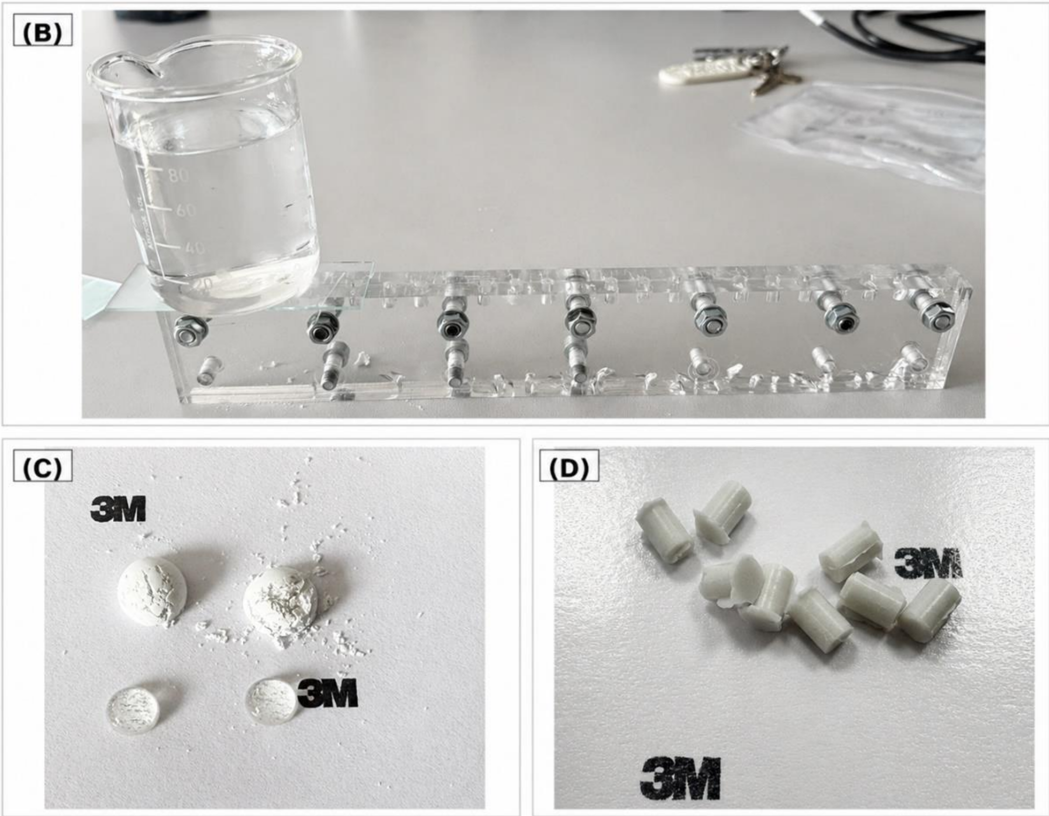


Figure 9: (A) Mould used for the cylindrical specimens, (B) Two scoops of glass powder and two drops of aqueous acid solution before mixing (3M is just the background and the acid solution, not the powder), (C) Specimens before polishing.

For the microhardness and surface roughness test, the samples created were disc-shaped samples. Their diameter was 10 mm and the thickness was 2 mm. The same samples were used by Łępicka et al [22]. It should be noted that a metal washer was used in the manufacture of the samples, as can be seen in the *Figure 10*.

***Figure 10**: Disc-shaped specimen*

4.3. Compressive strength test

4.3.1. Theoretical foundation

The compression test is used to determine the behaviour of a material subjected to crushing loads. The purpose of the compression test is to obtain a stress-strain diagram, from which other characteristics of a material are obtained, such as compressive strength, elastic limit, proportional limit, yield point, yield strength. To do this, a specimen with known and standardized dimensions, according to an international standard such as American Society of Testing and Materials (ASTM) or International Standards Organization (ISO), is subjected to a progressively increasing, uniformly distributed longitudinal load using specialized plates in a universal testing machine [42], [43], [44].

Not just any geometry is suitable for this type of test specimen. Cylindrical specimens are typically used with a height no greater than three times the diameter to prevent buckling [45].

Below, it is shown in *Figure 11* a typical stress-strain graph obtained in a compression test and explains its main characteristics [46]:

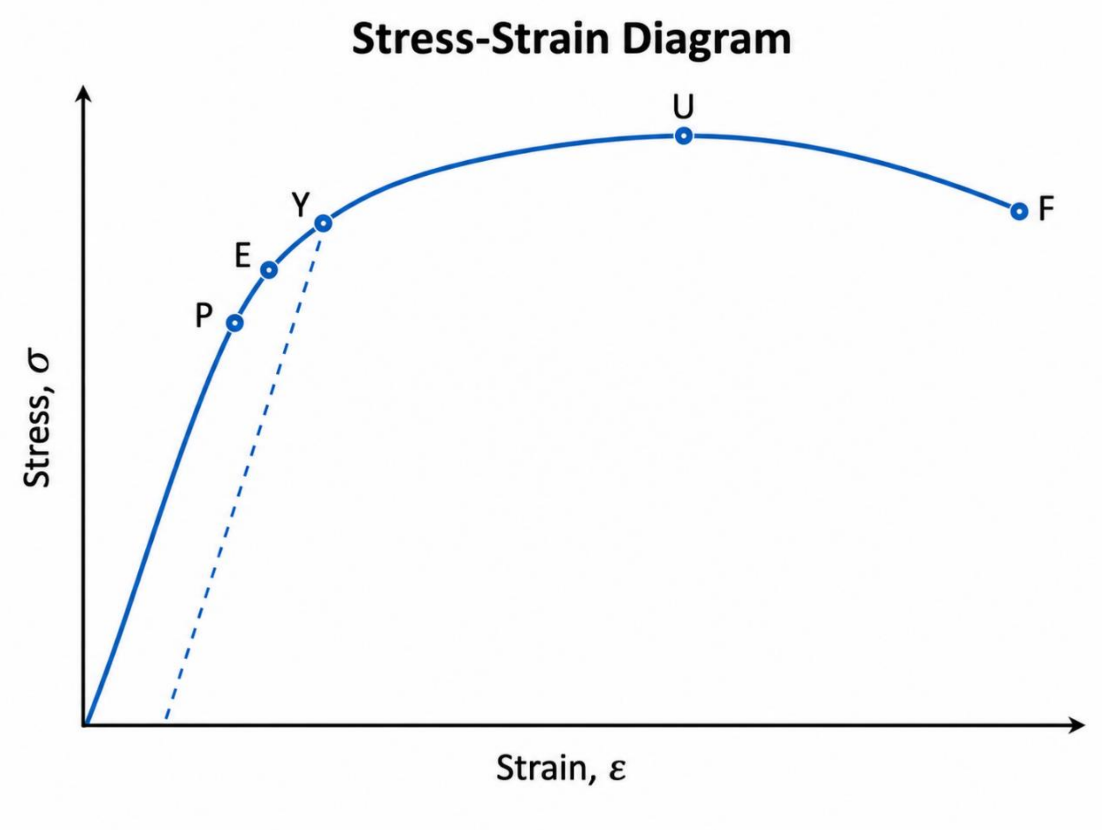


Figure 11: *Stress-strain curve obtained in compression test [46].*

Stress: $$\sigma = P/A_o \quad (1)$$

Strain: $$\varepsilon = \frac{\Delta L}{L_o} \quad (2)$$

In the stress equation, P represents the applied load, and A_o corresponds to the original cross-sectional area of the specimen. In the strain equation, ΔL is the change in specimen height during the compression test, whereas L_o is its original length.

Going with the points marked on the graph:

- P (*proportionality limit)*: This point marks the end of the initial linear part of the curve. Up to this point, stress and strain increase in direct proportion.
- E (*elastic limit)*: This point represents the highest stress that the material can withstand and still recover its original shape after removing the load. Between P and E, the curve may no longer be perfectly straight, but the deformation is still reversible.
- Y (*yield point)*: This point marks the beginning of permanent deformation. After this point, the material response becomes non-linear, and strain increases more rapidly with stress. If the yield point is not clearly visible, it can be estimated using the 0.2% offset method.
- U (*ultimate compressive strength)*: This point indicates the maximum stress reached by the material during the compression test.
- F (*fracture point)*: This is the final failure point. At this stage, the material can no longer support the load and breaks [46].

It is also necessary to explain the elastic modulus, which is the slope of the curve in the region of linear behaviour [46]:

Elastic modulus: $$E = \sigma/\varepsilon \quad (3)$$

It is also important to highlight the difference in the stress-strain curve obtained in a compression test for a ductile material and for a brittle one (*Figure 12*):

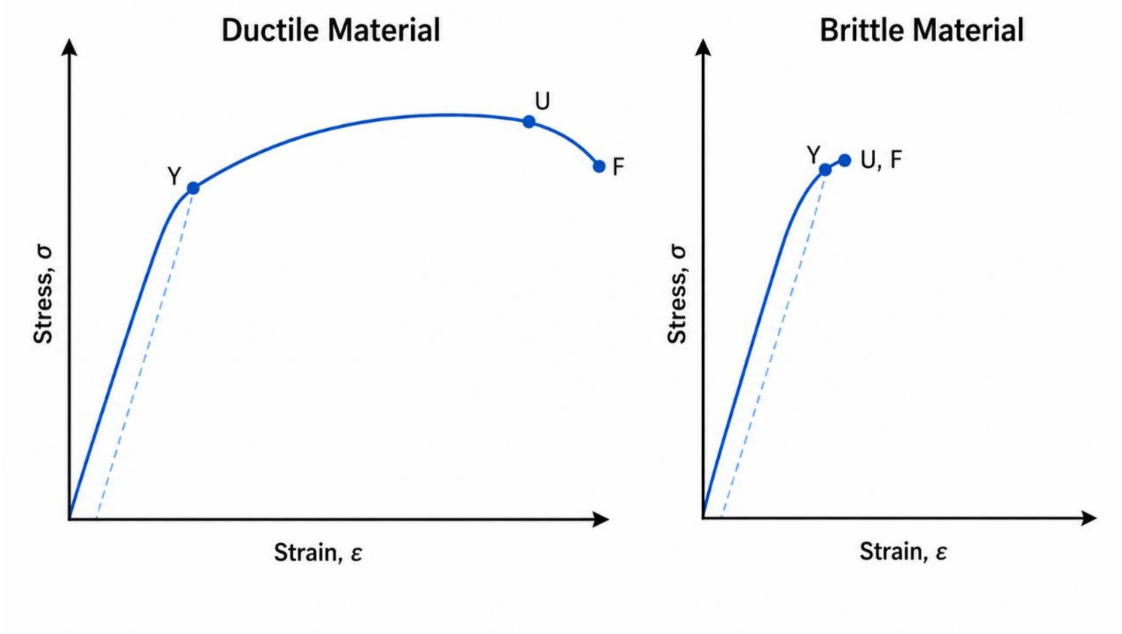


Figure 12*: Stress-strain curve comparison between ductile and brittle materials [46].*

It can be observed that for brittle materials, plastic deformation before fracture is non-existent (or minimal). In the case of glass-ionomer cements, the expected curve will be that of a brittle material, since it is a glass-based material.

4.3.2. Experimental process

Before mechanical testing, the dimensions of the specimens were measured using a digital micrometer. The height and diameter of each sample were determined in order to verify the final geometry of the specimens and to calculate the cross-sectional area required for the compressive strength calculations. Each specimen was measured immediately before being subjected to the corresponding mechanical test (*Figure 13*).

During the measurement procedure, special care was taken to minimize the time for which the specimens remained outside the storage medium. After measurement, the samples were promptly reimmersed in deionized water. This step was necessary because drying may weaken the structure of glass ionomer cement specimens and consequently affect their mechanical behaviour.

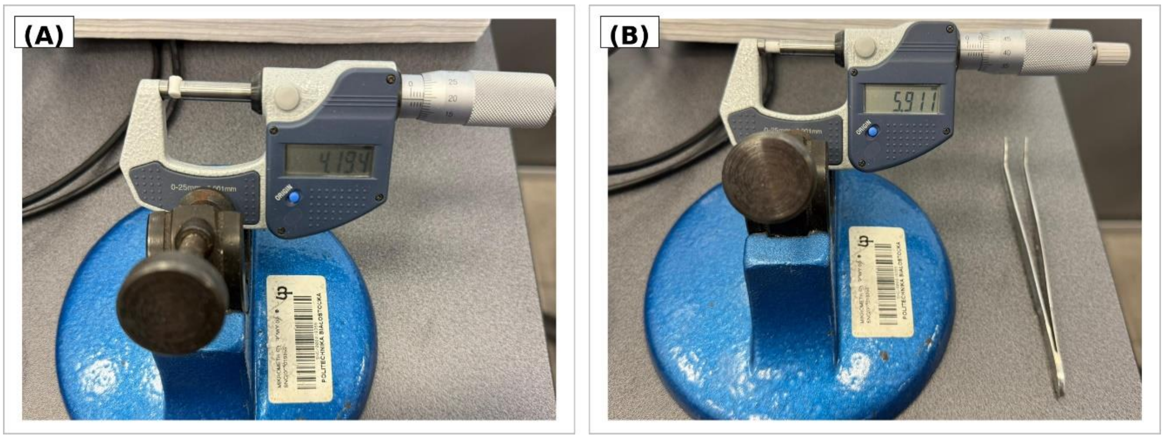


***Figure 13**: Dimensional measurement of the specimens before compression testing: (A) diameter measurement using a digital micrometer and (B) height measurement using a digital micrometer.*

The compressive strength test was performed using an MTS 322 dynamic testing machine (MTS Systems; Eden Prairie, USA). Before testing, the specimens were removed directly from deionized water and positioned in the centre of the hardened compression platens to ensure uniform load application along the longitudinal axis of the cylindrical sample.

For this test, 13 specimens from each of the four experimental groups were used: Sm0, Sm0.5, Sm1, and Sm5. The test procedure was carried out in accordance with ISO 9917-1. Each cylindrical specimen was first placed on the lower compression platen. A small gap was initially maintained between the specimen and the upper platen to allow accurate alignment of the sample with the loading axis. After positioning, the upper platen was lowered until it came into contact with the top surface of the specimen.

The specimens were then subjected to continuous axial compression until fracture. The test conditions were as follows: the sampling frequency was approximately 0.02 Hz, the force loading rate was approximately 0.5 N/s, and the crosshead speed was approximately 0.005 mm/s. The force-displacement data recorded during the test were saved for further analysis and used to determine the compressive strength of each specimen.

After testing, the fractured fragments were collected and stored for subsequent microscopic examination. The fracture surfaces were intended for further analysis using scanning electron microscopy in order to observe the internal structure of the material and the fracture morphology after mechanical failure.

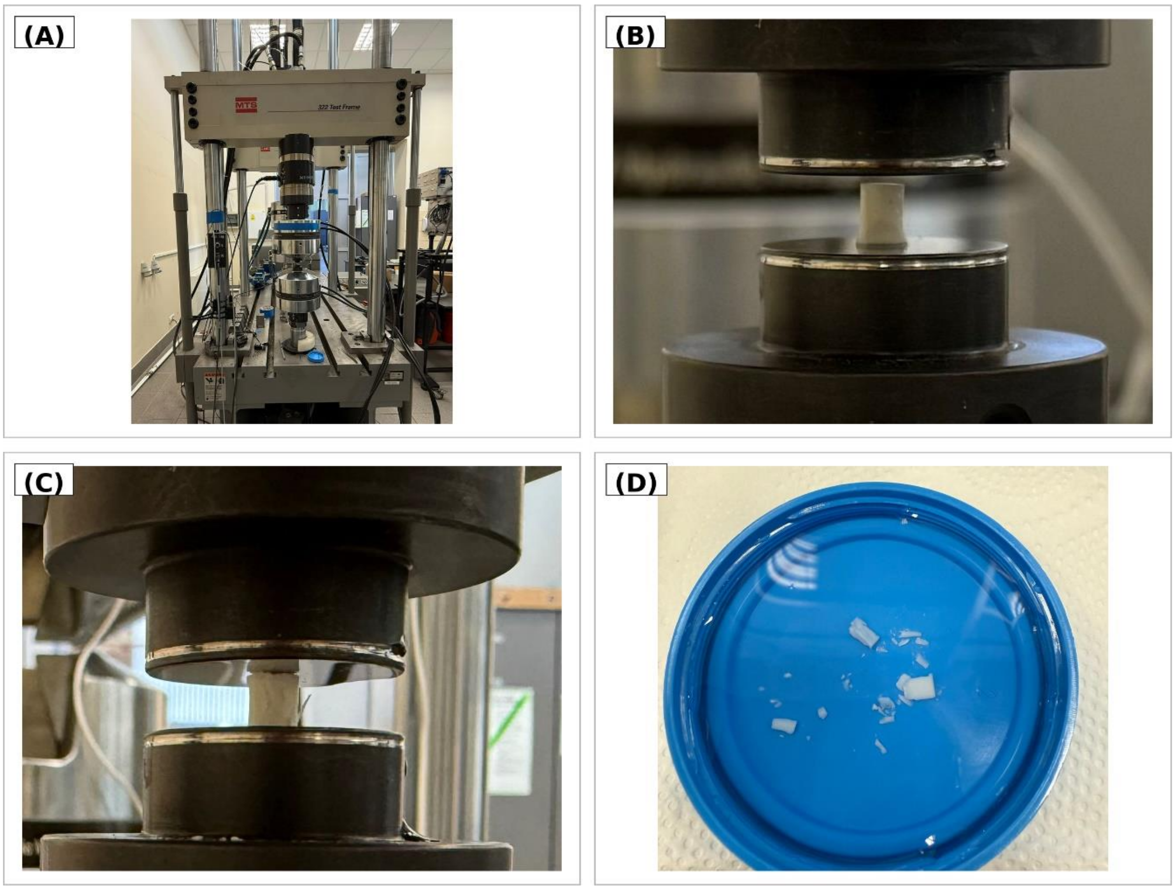


Figure 14: Compressive strength test procedure: (A) MTS 322 Test Frame universal testing machine, (B) specimen positioned between the compression plates before testing, (C) specimen during axial compression, and (D) fractured fragments after the test.

The compressive strength was calculated by dividing the maximum compressive force recorded during the test by the cross-sectional area of the specimen, according to the following equation:

$$R_c = F_c/A \tag{4}$$

where R_c is the compressive strength [MPa], F_c is the normal force recorded [N], and A is the cross-sectional area perpendicular to the loading direction [mm^2].

4.4. SEM microscopic observations

Samples were analysed using an SEM–FIB system, Scios 2 DualBeam SEM-FIB (Thermo Fisher Scientific; Waltham, USA) (*Figure 15*). The analysis was performed to observe the fracture morphology, the distribution of glass particles within the cement matrix and possible microstructural differences between the reference and samarium-doped groups.

Figure 15: Scios 2 DualBeam SEM–FIB system (Thermo Fisher Scientific; Waltham, USA) used for the microscopic analysis of fractured glass-ionomer cement specimens.

After the compressive strength test, fractured specimens were collected and stored for microscopic analysis. Representative fragments were selected from each experimental group: 0Sm, 0.5Sm, 1Sm, and 5Sm. The fragments were mounted on conductive carbon adhesive tape placed on metallic SEM stubs prior to observation.

The use of carbon adhesive tape allowed the fragments to be fixed securely and improved the electrical contact between the sample and the metallic support. This step was particularly important because glass ionomer cements are poorly conductive materials. During SEM observation, non-conductive specimens may accumulate electric charge on their surface, which can lead to image distortion, excessive brightness, and loss of microstructural details. Therefore, conductive mounting was used to reduce charging effects and improve image quality [47].

The observations were performed in a vacuum chamber to minimize interactions between the electron beam and air molecules [47]. An in-lens backscattered electron detector (T1) was used for imaging. Images obtained with this detector are based on backscattered electrons emitted from the analysed material. The contrast in backscattered electron images depends mainly on the atomic number of the elements present in the sample. Areas containing elements with higher atomic numbers appear brighter, whereas regions composed of elements with lower atomic numbers appear darker, because they backscatter fewer electrons [48]. This imaging mode was therefore useful for identifying differences in composition and morphology within the cement microstructure.

***Figure 16**: Samples prepared for SEM: particles of GICs used in the study*

4.5. Roughness test

4.5.1. Theoretical foundation

To measure surface roughness, a non-contact method using a confocal microscope belonging to the KEYENCE brand was employed due to its greater accuracy and faster measurement compared with contact methods. The contact methods are limited by the diameter of the needle, and they can also scratch the material REFERENCIA MAGDALENA.. This type of instrument uses a laser to measure the roughness of the analysed surface [49].

In the KEYENCE VK-X Series, the measurement unit includes an integrated X–Y scanner. During measurement, the laser scans the sample surface in both directions and collects the data needed to reconstruct the surface profile [49].

First, the laser moves across the surface in the X and Y directions and records the reflected light. Then, the objective lens moves slightly in the Z direction, and the scan is repeated at a different height. This process is repeated many times. For each point on the surface, the microscope keeps the height where the strongest reflected light is detected. In this way, the system can build a detailed 3D profile of the sample surface [49]. *Figure 17* shows the internal features of the microscope.

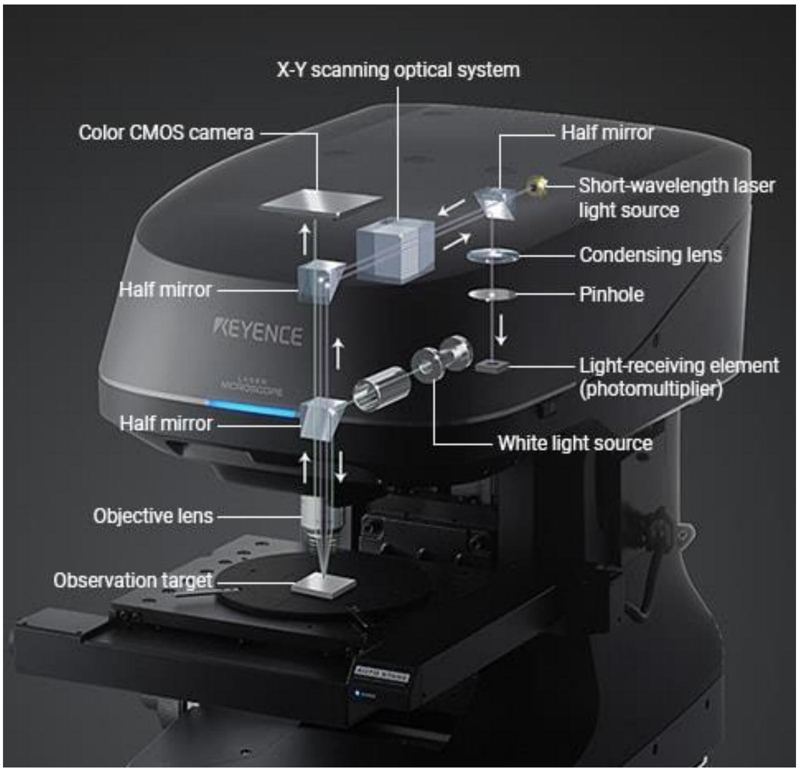


Figure 17: KEYENCE's VK-X Series 3D laser scanning microscope [49].

4.5.2. Experimental process

To carry out the surface roughness study of the samples, the Plan 10X objective lens of the KEYENCE VK-X Series 3D laser scanning microscope was used, with a working distance of 16.5 mm (*Figure 18*).

The samples used for this study were disc-shaped. One specimen from each of the four experimental groups was used: Sm0, Sm0.5, Sm1, and Sm5. Each sample was analysed on both sides (top and bottom).

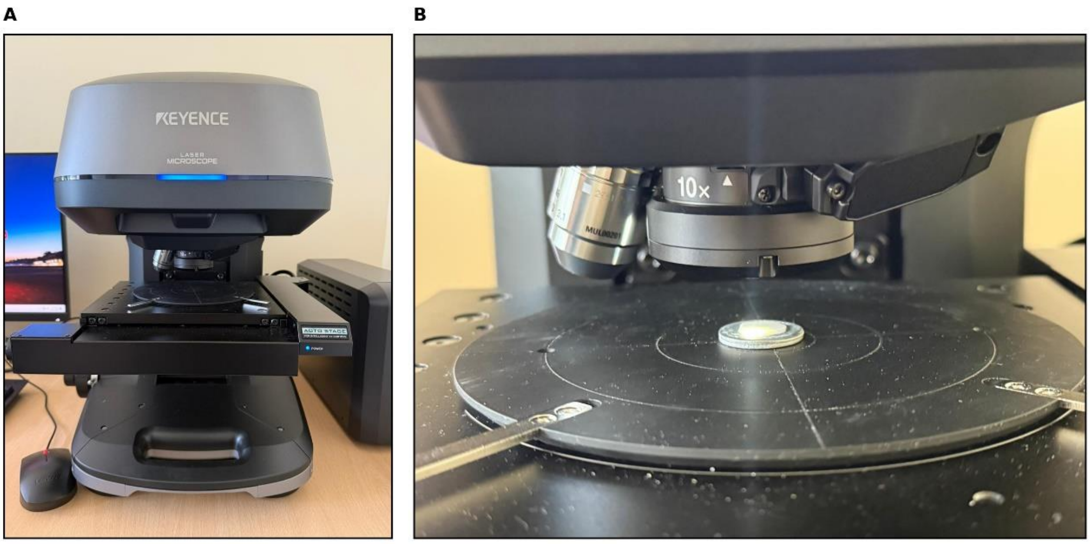
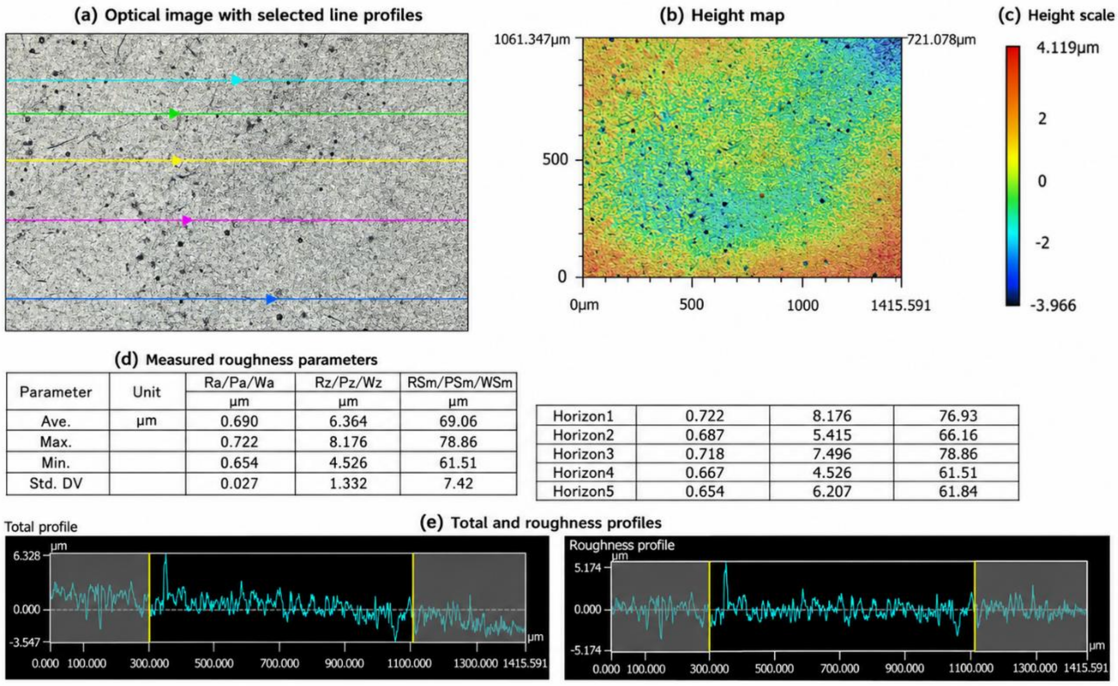


Figure 18: *Surface roughness measurement of glass ionomer cement samples using confocal microscopy. In Figure A, the confocal microscope used for the measurements is shown, while Figure B shows the positioning of the sample during the analysis.*

For each sample surface, laser images, 3D topographical reconstructions and height colour maps were obtained. To do this, five horizontal line profiles were selected on each surface, named Horizon 1 to Horizon 5. Then, the roughness parameters R_a, R_z and R_{Sm} were automatically calculated by the software in micrometres (*Figure 19*).



Parameter	Unit	Ra/Pa/Wa	Rz/Pz/Wz	RSm/PSm/WSm
		μm	μm	μm
Ave.	μm	0.690	6.364	69.06
Max.		0.722	8.176	78.86
Min.		0.654	4.526	61.51
Std. DV		0.027	1.332	7.42

Horizon1	0.722	8.176	76.93
Horizon2	0.687	5.415	66.16
Horizon3	0.718	7.496	78.86
Horizon4	0.667	4.526	61.51
Horizon5	0.654	6.207	61.84

Figure 19: *Representative example of the output generated by the KEYENCE VK-X 3D laser scanning microscope software. Line roughness analysis of the surface, including the optical image with the selected*

profiles, the height map, the colour scale, the measured roughness parameters, and the extracted total and roughness profiles.

4.6. Microhardness test

4.6.1. Theoretical foundation

Hardness is the ability of a material to resist permanent deformation caused by the indentation of a rigid body [50]. Hardness tests are performed to study the scratch and wear resistance of materials [45]. The test used in this study is the Vickers hardness test, which was developed in 1922. This test consists of making an indentation with a pyramidal diamond indenter whose opposite faces form 136º [50]. The applied load ranges from 1 kg to 20 kg and is applied for between 10 and 15 seconds. In this test, the most important data point is the length of the indentation diagonals (*Figure 20*), which are usually less than 0.5 mm and are measured with a microscope. Finally, the microhardness result is measured in HV [45], [50].

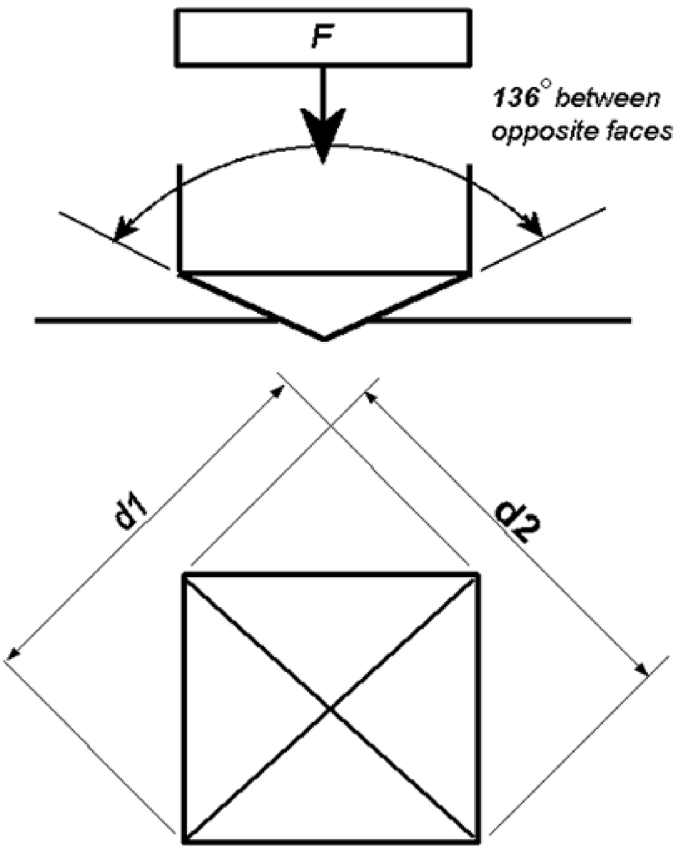


***Figure 20**: Vickers indentation [50].*

The value of the Vickers hardness is given by the following expression:

$$HV = \frac{2Fsin(\frac{136º}{2})}{d^2} \qquad (5)$$

were *HV* is the Vickers hardness [N/mm^2], *F* is the normal force applied [kgf], *d* is the mean value of the measured diagonals [mm] [50].

4.6.2. Experimental process

Surface microhardness measurements were performed using a microhardness tester MicVision VH-1 (Sinowon; Dong Guan, China) shown in *Figure 21*:

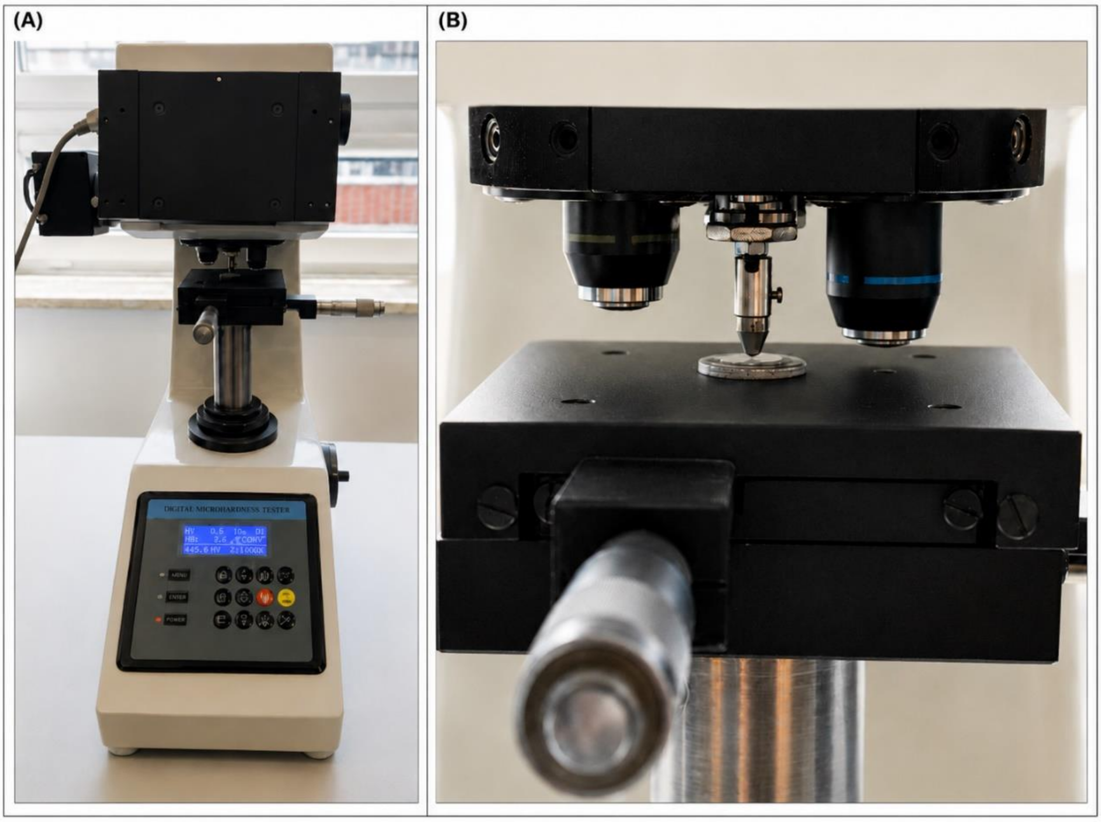


Figure 21*: Digital microhardness tester used for Vickers microhardness measurements: (A) general view of the equipment; (B) detail of the indenter and sample positioning during the test.*

During the test, a load of 1000 g was applied, corresponding to a normal force of 9.81 N. After indentation, the diagonals of the imprint left on the specimen surface were measured. These values were then used to calculate the Vickers microhardness of the tested glass ionomer cement specimens.

For each experimental group, 15 measurements were performed. The obtained indentation images allowed the diagonal lengths to be determined and used for further calculation of the surface microhardness values. An example of an indentation image with the measured diagonals is shown in *Figure 22*.

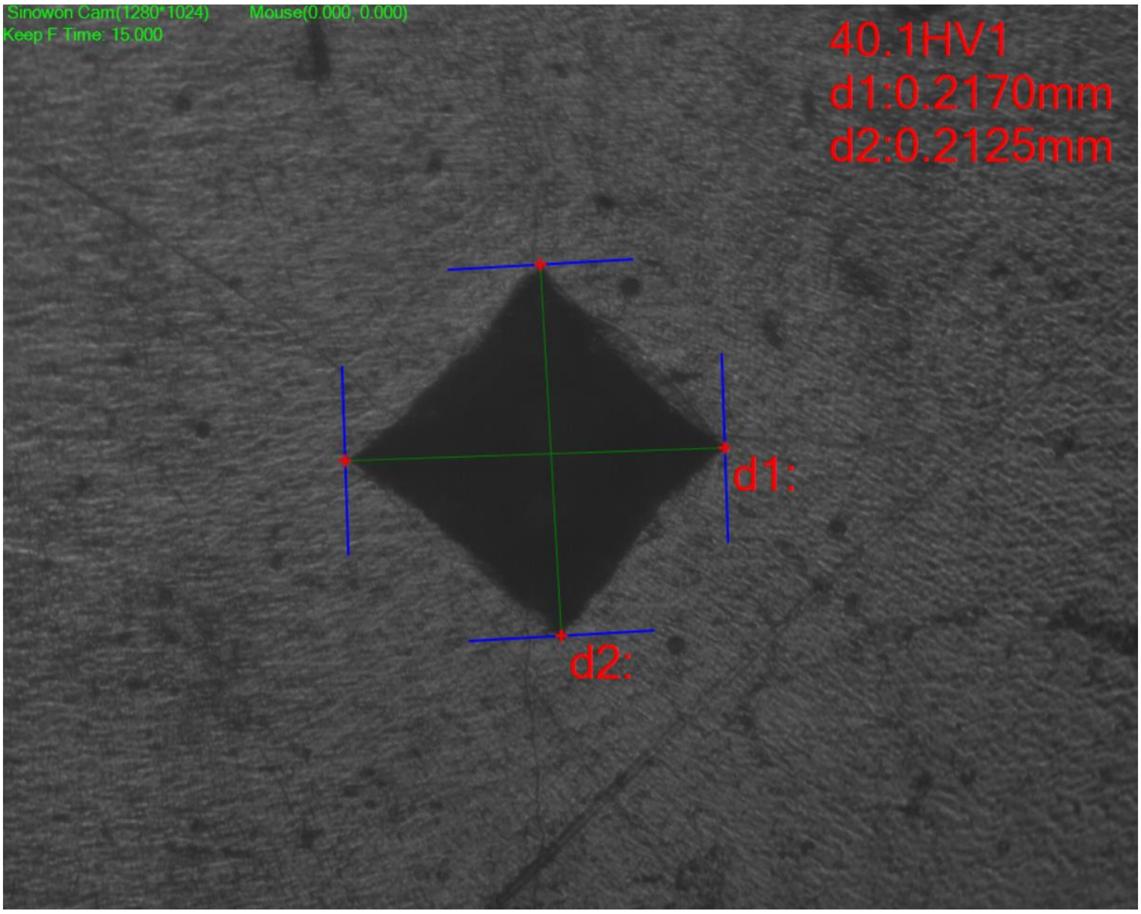


***Figure 22**: Vickers test: diagonals measurement*

The equipment used in the laboratory directly provided the Vickers hardness reading in the screen.

4.7. Acid pretreatment of glass powder with hydrochloric acid

Following the compression test performed on samples of samarium-treated glass ionomer cement, it was found that the compressive strength was insufficient. Therefore, a solution was proposed which consisted of pretreating the glass with hydrochloric acid (HCl) before mixing it with the aqueous acid solution.

The purpose of this treatment was to clean the surface of the glass particles, remove possible residues produced during the crushing process and produce a mild surface activation. The intention was to modify the surface of the powder slightly. This could improve the interaction between the glass particles and the polyacid matrix during the setting reaction of the cement.

Materials and Equipment

Materials:

- Glass powder: 2.000 g
- Concentrated HCl: 35–38 wt%
- Deionized water

Equipment:

- Ultrasonic probe
- Laboratory centrifuge
- PP/PTFE or other HCl-resistant vessel
- HCl-resistant centrifuge tubes
- Laboratory oven
- Agate mortar and pestle or another clean manual grinding set
- pH paper or pH meter
- Desiccator or sealed dry container

Preparation of 5 wt% hydrochloric acid solution

Before treating the glass powder, a hydrochloric acid solution of approximately 5 wt% HCl was prepared. The concentrated hydrochloric acid available in the laboratory had a concentration of 37 wt% HCl. Therefore, 5.7 mL of concentrated HCl were used to prepare 50 mL of diluted acid solution.

First, approximately 35–40 mL of deionized water were added into a PP/PTFE acid-resistant vessel. Then, 5.7 mL of concentrated HCl were slowly added to the water, since it is an exothermic reaction and sufficient time must be given for the heat to dissipate gradually. Finally, deionized water was added until reaching a final volume of 50 mL.

During this step, the acid was always added to water, and not water to acid, in order to reduce the risk of a strong exothermic reaction.

HCl washing using an ultrasonic probe

For the acid treatment, 2.000 g of glass powder were weighed using a precision laboratory balance. The powder was then transferred into a PP/PTFE HCl-resistant vessel. After this, 40 mL of freshly prepared 5 wt% HCl solution were added to the glass powder.

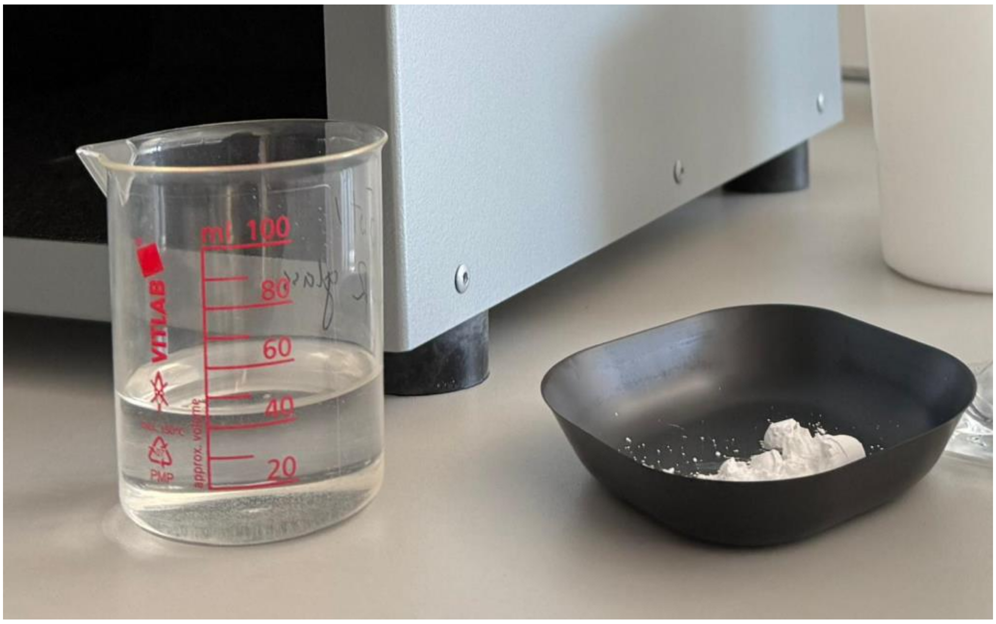


***Figure 23**: Glass powder and freshly prepared 5 wt% HCl solution before mixing*

Immediately after adding the acid solution, the tip of the ultrasonic probe was immersed into the suspension. The probe tip was placed below the liquid surface, but it was kept away from the bottom and the walls of the vessel.

Figure 24: Ultrasonic probe immersed in the glass powder/HCl suspension during the acid washing step

The suspension was sonicated using a pulsed mode consisting of 3 rounds of 15 seconds each with 5 s pauses. This means that the total sonication time was 45 s. However, the total acid contact time was approximately 60 s, counted from the moment when the HCl solution was added to the glass powder until the treatment was stopped by dilution with deionized water.

The ultrasonic probe was used under low to moderate intensity conditions, with an amplitude of approximately 20–30%. These conditions were selected to avoid overheating of the suspension and to prevent excessive chemical attack on the surface of the glass particles.

Immediate dilution and reaction stop

Immediately after the final ultrasonic pulse, the acid treatment was stopped by diluting 80–100 mL of deionized water added directly to the suspension. The mixture was then briefly mixed for approximately 10–20 s.

This dilution step was important because it reduced the acid concentration and stopped the surface treatment of the glass powder. After dilution, the suspension was transferred into HCl-resistant centrifuge tubes for the washing stage.

Centrifugation and washing

The treated glass powder was separated from the acidic liquid by centrifugation. The suspension was centrifuged at approximately 4000-6000 rpm for 10 min.

Figure 25*: Centrifugation machine*

After centrifugation, the acidic supernatant was carefully removed. Subsequently, fresh deionized water was then added to the powder. Approximately 40–50 mL of deionized water was used for each washing step. The powder was redispersed by manually shaking. Then, the suspension was centrifuged again under the same conditions, at 4000-6000 rpm for 5-10 min. This washing cycle was repeated at least four times in order to remove the remaining acid from the powder. To verify this, pH paper was used, which showed that the supernatant reached an approximately neutral pH, between pH 5.5 and 6.5.

Drying of the treated glass powder

After the final washing step, the wet glass powder was collected and transferred onto a clean ceramic drying dish. The powder was then dried in a laboratory oven at 105 °C for 48 h.

***Figure 26**: Laboratory oven*

Once the drying process was completed, the powder was removed from the oven. Then, it was gently deagglomerated using a spatula. This step was carried out only to break the soft agglomerates and obtain a free-flowing powder. Intensive grinding was avoided because the aim was not to significantly reduce the particle size. The purpose was only to improve the handling and homogeneity of the powder before preparing the cement samples. Nevertheless, the particle size was quite small.

After deagglomeration, the treated powder was stored in a tightly sealed container under dry conditions. This powder was later used for the preparation of new glass-ionomer cement specimens, following the same cement preparation procedure used for the untreated glass powder.

4.8. Sample preparation of HCl-treated glass-ionomer cement specimens

After the acid pretreatment of the glass powder, new glass-ionomer cement specimens were prepared for compressive strength testing. The objective was to evaluate whether the HCl treatment of the glass powder could improve the mechanical behaviour of the cement.

Two types of HCl-treated powders were used for this preliminary preparation: a reference glass powder without samarium and a glass powder containing 5 mol% Sm_2O_3. In both cases, the powders were previously treated with hydrochloric acid following the procedure described above.

The cement specimens were prepared using the same procedure previously used for the untreated glass powders. The powder and liquid were mixed until a homogeneous paste was obtained, then the paste was introduced into the cylindrical moulds. The moulds were covered and pressed in the same way as in the previous sample preparation process. After the initial setting time, the specimens were removed from the moulds and placed in deionized water in an oven at 37ºC for a week.

The specimens prepared with the HCl-treated glass powder without samarium maintained their shape after demoulding. However, the specimens prepared with the HCl-treated glass powder containing 5 mol% Sm_2O_3 showed poor stability after being placed in deionized water (*Figure 27*). Shortly after immersion, these samples started to loose their structural integrity and began to disintegrate in the water.

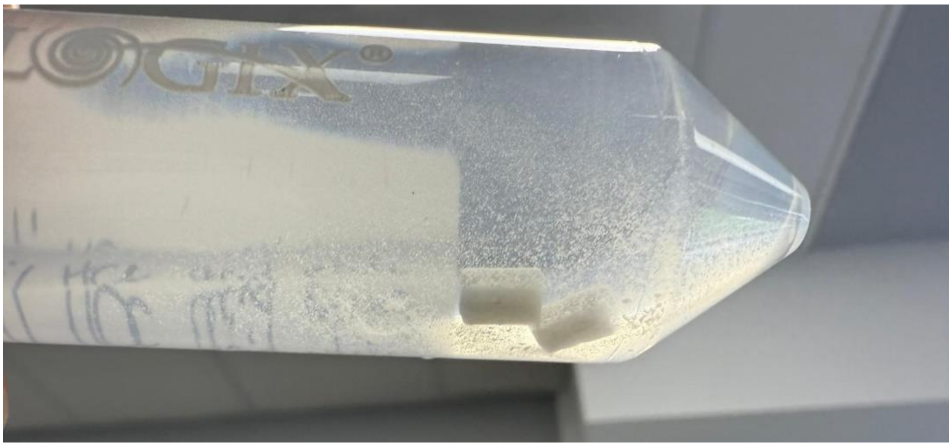


Figure 27: *Specimens prepared with the HCL-treated glass powder containing 5 mol% Sm_2O_3*

5. Results

Statistical analysis was performed using Statistica software (TIBCO Software Inc.). The analysed groups included the reference material without samarium addition (Sm0) and the samarium-modified materials containing 0.5, 1 and 5 wt.% Sm.

5.1. Compressive strength test results

The data obtained from the compression tests were recorded by the software connected to the testing machine and processed afterwards. To obtain the mechanical response of each specimen, load-displacement data were used. Moreover, the maximum force at fracture was used to calculate the compressive strength.

After analyzing the data collected, the following graphs were obtained (*Figure 28 and 29*):

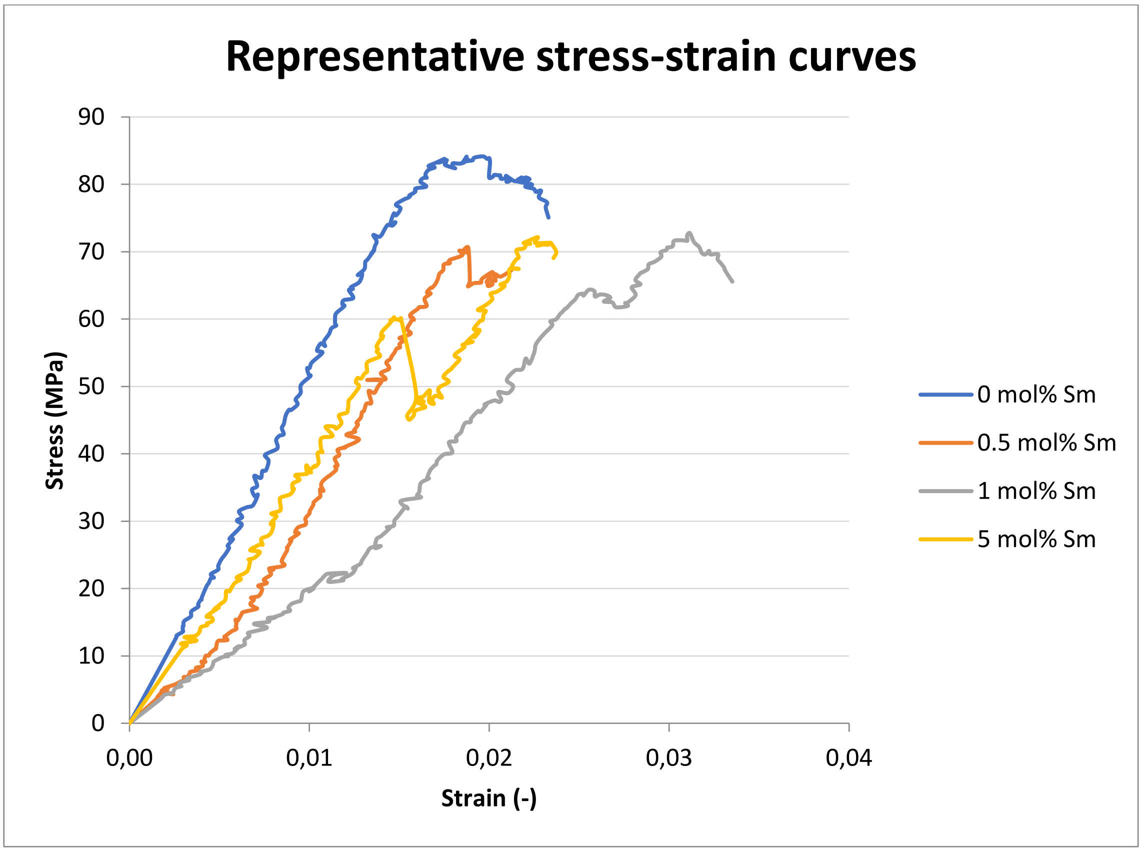


***Figure 28**: Representative stress-strain curves for each group*

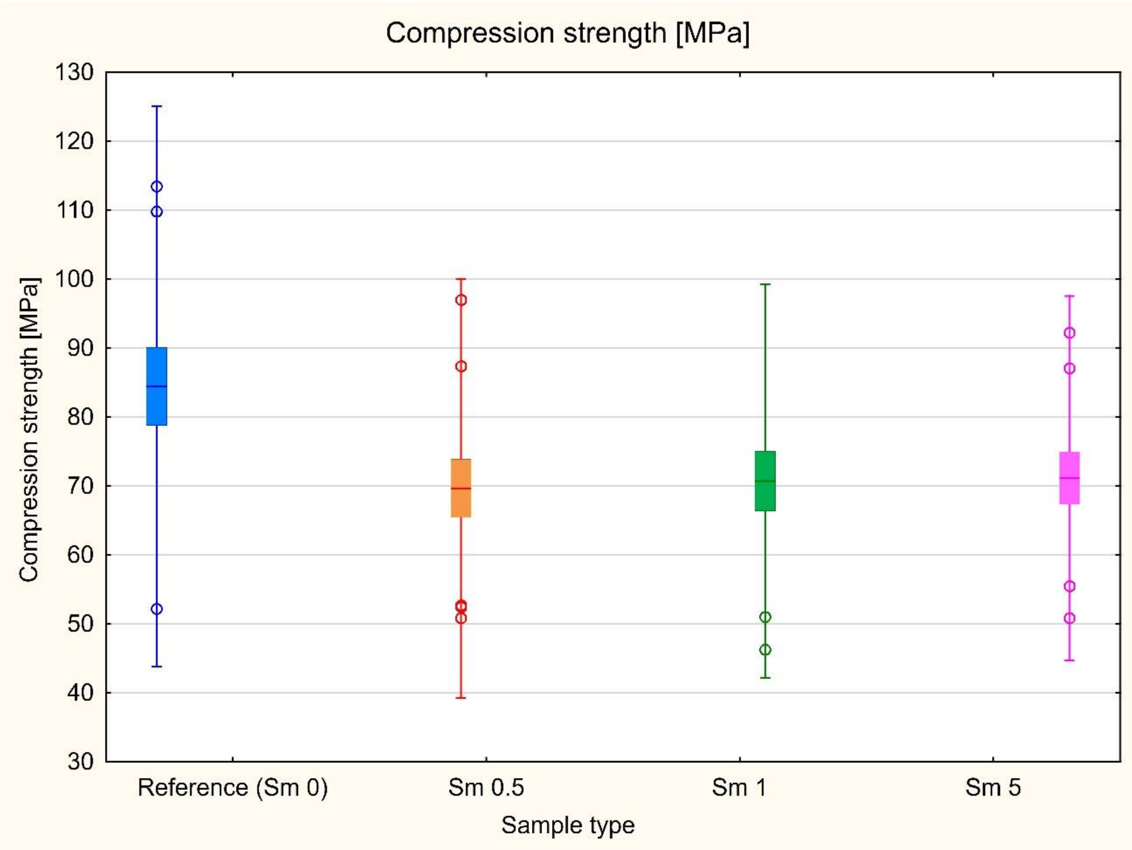


***Figure 29**: Compression test statistical analysis*

The normality of data distribution in each group was verified using the Shapiro-Wilk test. The obtained p-values ranged from 0.229 to 0.751, indicating that the assumption of normal distribution was met for all analysed groups. The homogeneity of variances was also confirmed by Levene's test ($p = 0.168$). This test allowed the use of one-way analysis of variance.

Regarding the ANOVA results, no statistically significant differences were found in compressive strength between the reference cement and the Sm-modified materials ($p = 0.079$). The samples that contained samarium showed lower mean compressive strength values in comparison with the reference material. Neverthelss, this difference did not reach statistical significance. These results indicate that samarium modification carried out in the investigated concentration range, did not significantly affect the compressive strength of the tested glass ionomer-based materials.

It should also be noted that the samples tested in each group showed great variability, possibly due to the presence of voids. This was observed in all four sample groups, so only one is shown below as an example (*Figure 30*).

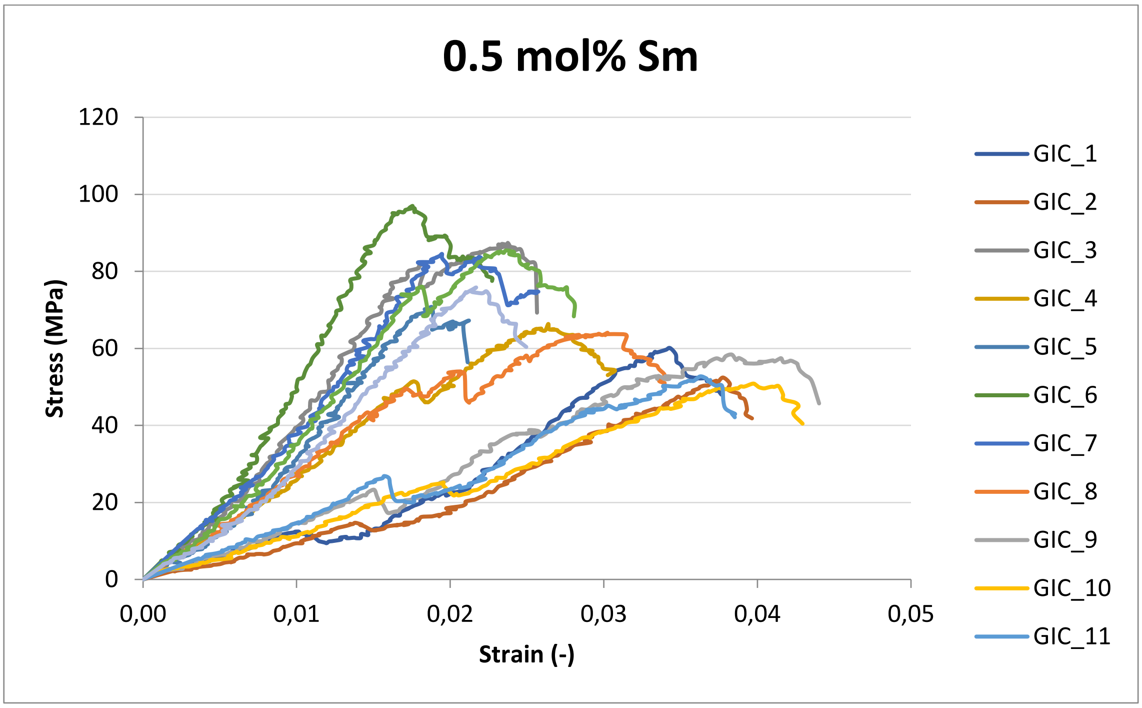


***Figure 30**: Stress-strain curve group for the 0.5%Sm group*

HCl pretreated samples

Regarding the samples previously treated with HCl in an attempt to improve compressive strength, the following results were obtained (*Figure 31 and 32*):

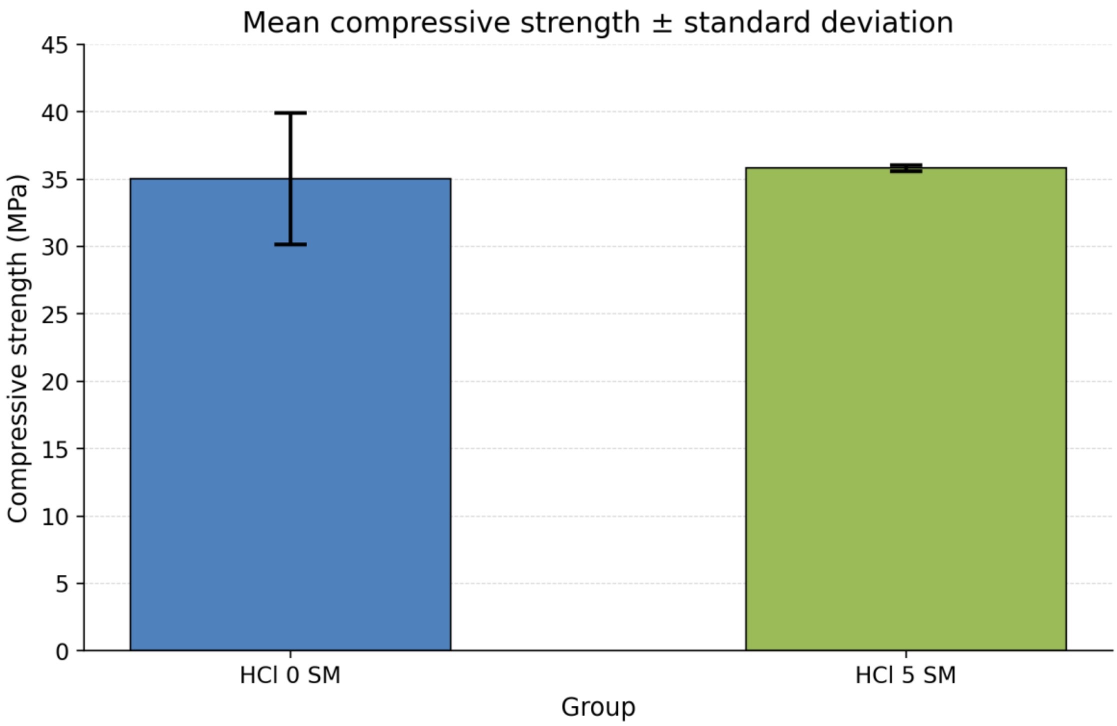


***Figure 31**: Compressive strength test results for HCl treated samples*

As can be observed, both samples not only failed to improve compressive strength but actually worsened considerably. The average values obtained were about 35 MPa, thus demonstrating that the process followed did not work as an improvement.

5.2. Roughness test results

Surface roughness - R_a

After analyzing the data collected, the following graphs were obtained (*Figure 32):*

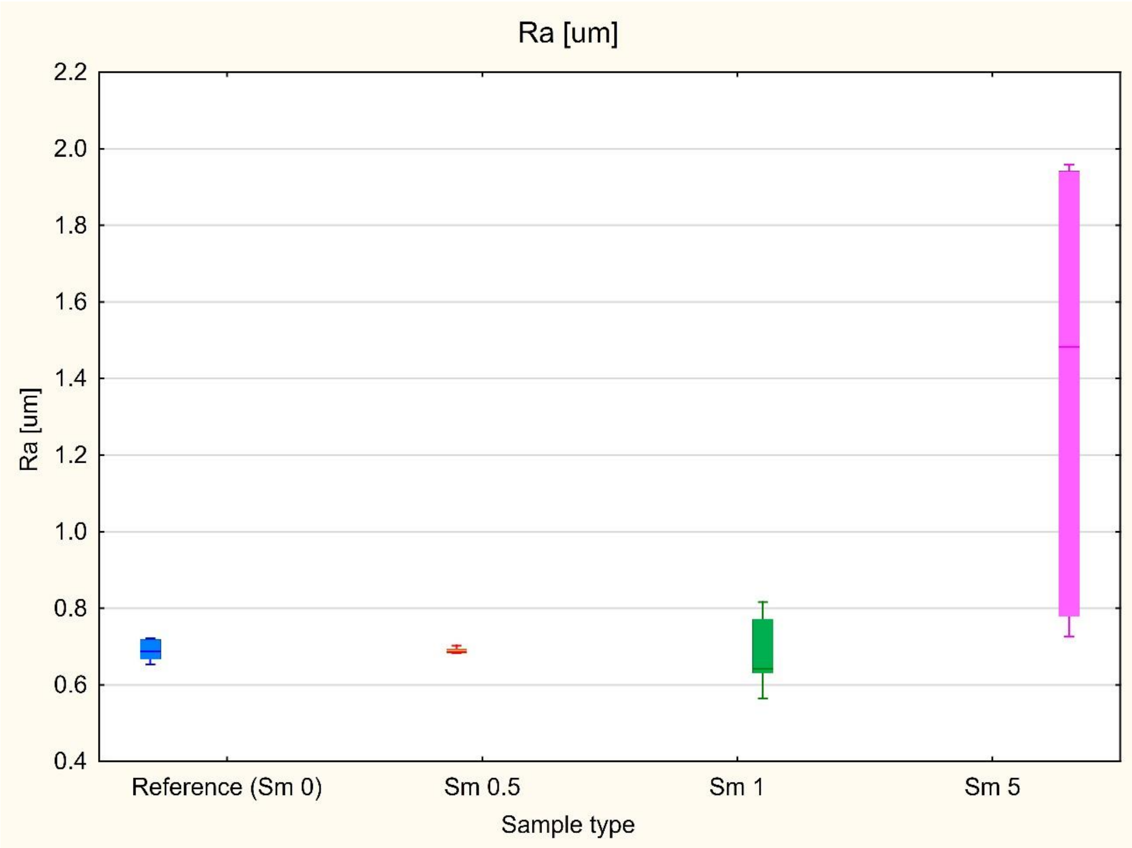


***Figure 32**: Roughness test statistical analysis*

Shapiro-Wilk test indicated that the assumption of normal distribution was not fulfilled for all groups, therefore, non-parametric analysis was applied.

One extreme outlier was removed, then the Kruskal-Wallis test revealed a statistically significant overall difference in R_a values between the analysed materials, $H(3, N = 19) = 8.56, p = 0.031$. Consequently, this indicates that material composition had a significant effect on surface roughness. Sm5 group showed the highest R_a values based on the mean ranks. The reference material, Sm0.5 and Sm1 exhibited lower and more comparable roughness levels.

The median test also confirmed significant differences between the groups, $\chi^2(3) = 9.37$, $p = 0.025$. In the Sm5 group, all observations were above the overall median. However, in the remaining groups the values were distributed around or below the median. Therefore, this suggests that material containing 5 wt.% Sm tends to have greater surface roughness.

However, post-hoc multiple comparisons (*Figure 33*) did not reveal statistically significant differences between any specific pair of groups after correction. The lowest pairwise p-value was observed for the comparison between Sm1 and Sm5, $p = 0.058$. This indicates a near-significant tendency towards higher Ra values in the Sm5 group. Therefore, although the global non-parametric tests suggest that material composition affects surface roughness, the pairwise differences should be interpreted with caution.

A

ANOVA rang Kruskala-Wallisa; Ra [um] (Roughness)
Zmienna niezależna (grupująca): Sample type
Test Kruskala-Wallisa: H (3, N= 19) =8.856316 p =.0313

Zależna: Ra [um]	Kod	N ważnych	Suma Rang	Średnia Ranga
Reference (Sm 0)	1	5	41.00000	8.20000
Sm 0.5	2	4	31.00000	7.75000
Sm 1	3	5	36.00000	7.20000
Sm 5	4	5	82.00000	16.40000

B

Test mediany, ogólna mediana= .702000; Ra [um] (Roughness)
Zmienna niezależna (grupująca): Sample type
Chi kwadrat= 9.373333 df = 3 p = .0247

Zależna: Ra [um]	Reference (Sm 0)	Sm 0.5	Sm 1	Sm 5	Razem
<=mediany:obserwow.	3.000000	4.00000	3.000000	0.00000	10.00000
oczekiwane	2.631579	2.10526	2.631579	2.63158	
obs.-ocz.	0.368421	1.89474	0.368421	-2.63158	
>mediany:obserwow.	2.000000	0.00000	2.000000	5.00000	9.00000
oczekiwane	2.368421	1.89474	2.368421	2.36842	
obs.-ocz.	-0.368421	-1.89474	-0.368421	2.63158	
Razem: obserwowane	5.000000	4.00000	5.000000	5.00000	19.00000

C

Wartość 'z' dla porównań wielokrotnych; Ra [um] (Roughness)
Zmienna niezależna (grupująca): Sample type
Test Kruskala-Wallisa: H (3, N= 19) =8.856316 p =.0313

Zależna: Ra [um]	Reference (Sm 0) R:8.2000	Sm 0.5 R:7.7500	Sm 1 R:7.2000	Sm 5 R:16.400
Reference (Sm 0)		0.119208	0.280976	2.304001
Sm 0.5	0.119208		0.145699	2.291441
Sm 1	0.280976	0.145699		2.584977
Sm 5	2.304001	2.291441	2.584977	

D

Wartość p dla porównań wielokrotnych (dwustronych); Ra [um] (Roughness)
Zmienna niezależna (grupująca): Sample type
Test Kruskala-Wallisa: H (3, N= 19) =8.856316 p =.0313

Zależna: Ra [um]	Reference (Sm 0) R:8.2000	Sm 0.5 R:7.7500	Sm 1 R:7.2000	Sm 5 R:16.400
Reference (Sm 0)		1.000000	1.000000	0.127335
Sm 0.5	1.000000		1.000000	0.131628
Sm 1	1.000000	1.000000		0.058431
Sm 5	0.127335	0.131628	0.058431	

***Figure 33**: Post-hoc analysis for roughness*

5.3. Microhardness test results

After analyzing the data collected, the following graphs were obtained (*Figure 34*):

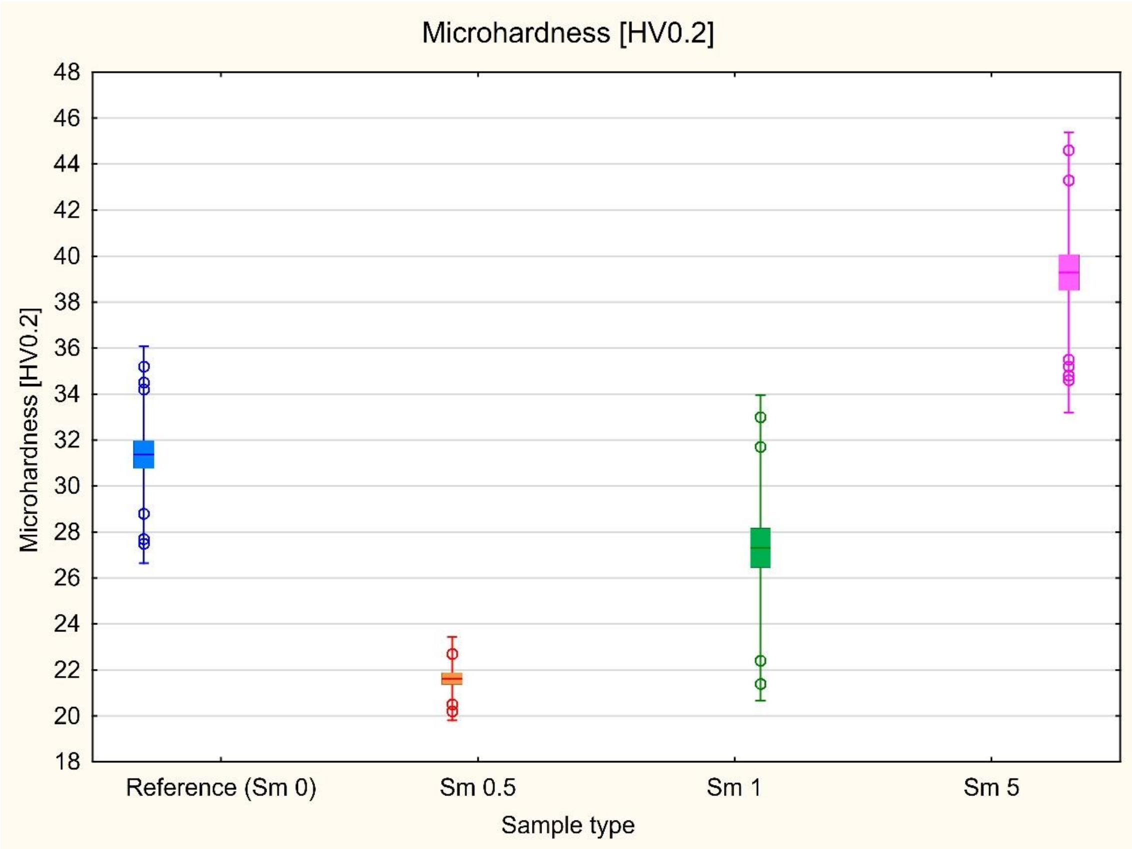


***Figure 34**: Microhardness test statistical analysis*

The Shapiro-Wilk test was used to verify the normality of data distribution in each group. The obtained p-values ranged from 0.119 to 0.965, indicating that the assumption of normal distribution was met for all analysed groups. However, the homogeneity of variances was not confirmed, as shown by Levene's test ($p = 0.008$). Therefore, analysis of variance was performed using Welch's correction. The Welch-corrected ANOVA revealed a statistically significant effect of material composition on microhardness ($p < 0.001$).

Post-hoc analysis using Tukey's RIR test showed statistically significant differences between all analysed material groups (*Figure 35*). The lowest microhardness was observed for the Sm0.5 material, followed by Sm1 and the reference material. The highest mean microhardness value was obtained for the Sm5 composition. These results indicate that samarium modification significantly affected the microhardness of the investigated glass ionomer-based materials, although the observed effect was not linear across the tested compositions.

	HSD (nierówne N); zmn.: Microhardness [HV0.2] (Microhardness) Zaznaczone różnice są istotne z p < .05000			
Sample type	**{1} M=31.367**	{2} M=21.625	{3} M=27.307	{4} M=39.287
Reference (Sm 0) {1}		0.000162	0.000708	0.000162
Sm 0.5 {2}	0.000162		0.000174	0.000162
Sm 1 {3}	0.000708	0.000174		0.000162
Sm 5 {4}	0.000162	0.000162	0.000162	

***Figure 35**: Post-hoc analysis*

5.4. SEM microscopic observations results

The images obtained in the SEM microscopy study for each of the four experimental groups are shown and analysed in this section.

This is the typical microstructure of a glass ionomer cement (*Figure 36*). We can see the hydrogel matrix (dark gray) which was formed during the cement setting. Moreover, particles of the non-reacted glass, visible as light-grey particles, are also present. Signs of particles dissolution are also seen. Some cracks appeared in the sample. However, they are the artifacts which are being formed during cement drying, as the cement needs to be completely dry when put in vacuum. Moreover, voids are present in the material. Most probably, those were formed during hand spatulation and agglutination of the cement substrates.

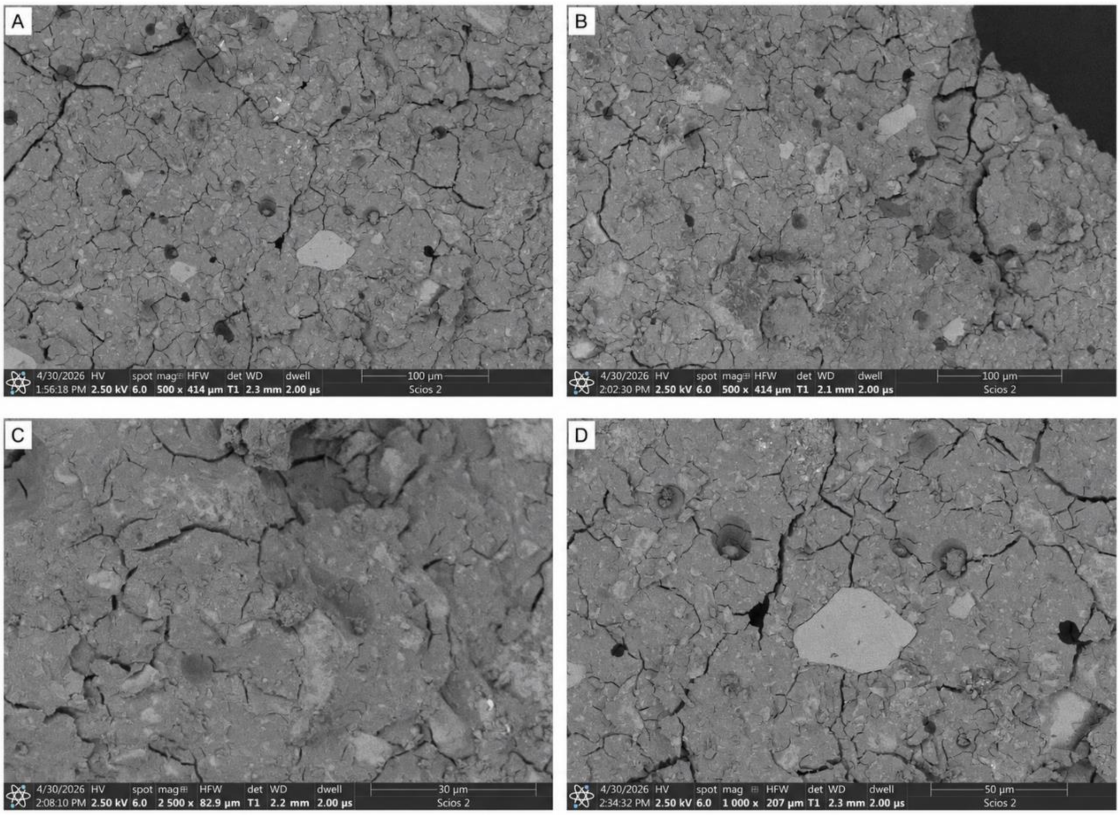


***Figure 36**: SEM images of the 0SM glass ionomer cement sample at different magnifications: (A) 500×, (B) 500×, (C) 2500×, and (D) 1000×.*

As can be seen, the microstructure does changes slightly when samarium is introduced in the matrix (*Figures 37, 38 and 39*). The more samarium is there, the more intact, non-reacted glass particles are visible. This might mean that the kinetics of the setting reaction changes when samarium is present in the material, making the glass less susceptible to acidic attack and the following ion exchange. Moreover, similarly to the reference sample, cracks, as well as voids, are seen.

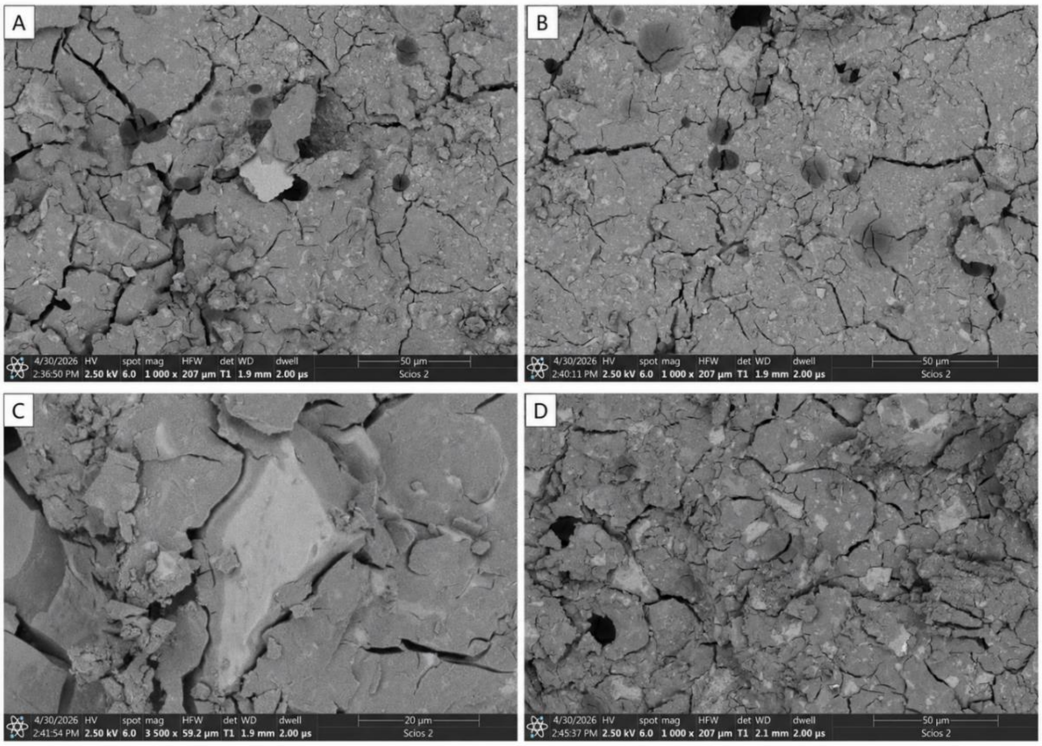


***Figure 37**: SEM images of the 0.5SM glass ionomer cement sample at different magnifications: (A) 1000×, (B) 1000×, (C) 3500×, and (D) 1000×.*

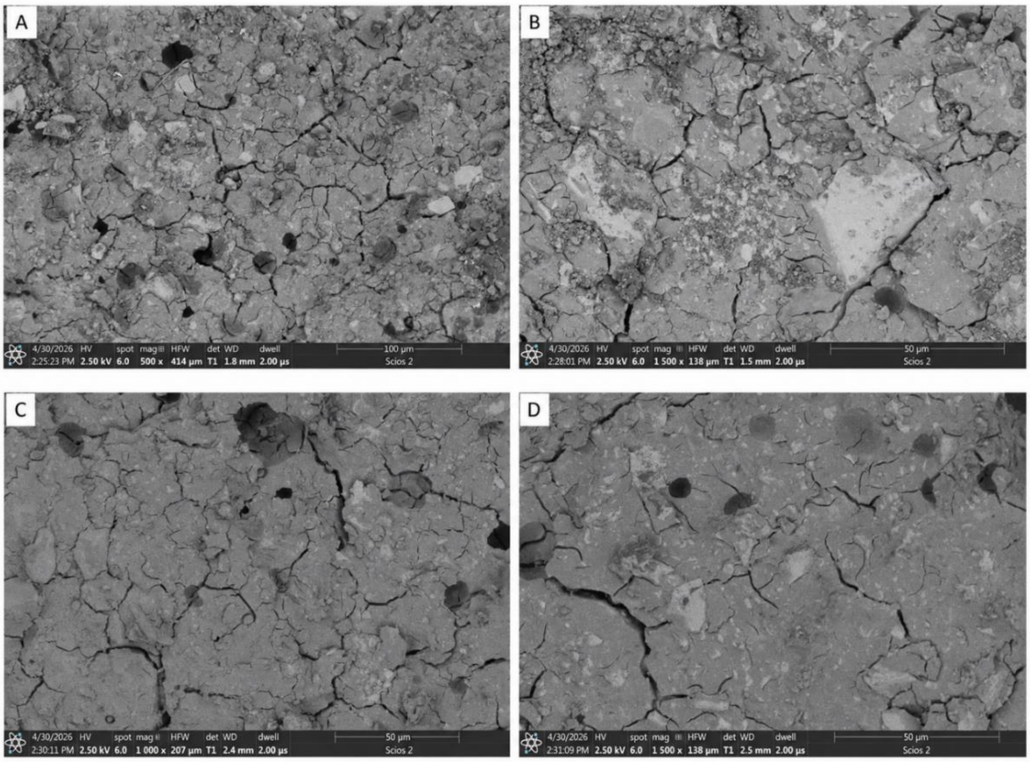


***Figure 38**: SEM images of the 1SM glass ionomer cement sample at different magnifications: (A) 500×, (B) 1500×, (C) 1000×, and (D) 1500×.*

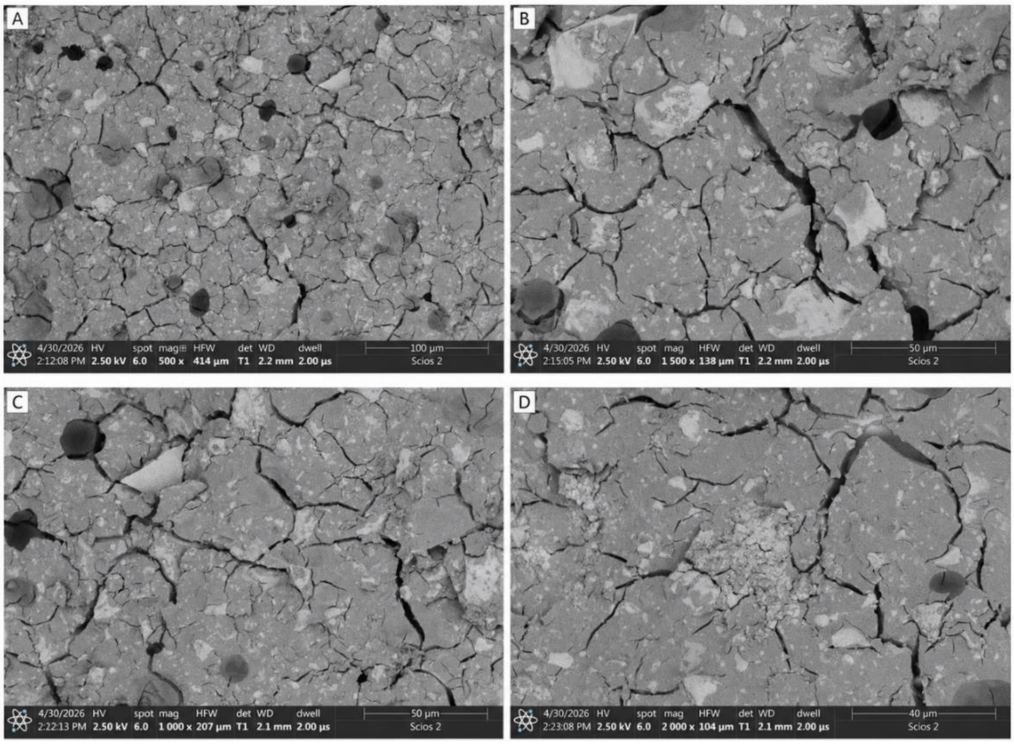


***Figure 39**: SEM images of the 5SM glass ionomer cement sample at different magnifications: (A) 500×, (B) 1500×, (C) 1000×, and (D) 2000×.*

6. Discussion

The aim of this study was to examine how the incorporation of samarium oxide affects the functional behaviour of experimental glass-ionomer cements. For this purpose, part of the SiO_2 in the glass phase was replaced by Sm_2O_3. Four compositions were prepared and analysed: one reference cement without samarium and three modified cements containing 0.5, 1 and 5 mol% Sm_2O_3. This approach is justified by the fact that samarium has already shown promising behaviour in hydroxyapatite, coatings, bioactive glasses and cement-like dental materials, whereas its influence on conventional glass-ionomer cement systems is still insufficiently explored. Therefore, the present work addresses an important research gap by assessing whether Sm_2O_3 can act as a functional modifier of glass-ionomer cement properties.

From a practical point of view, compressive strength is especially relevant in this type of material, because a dental cement has to tolerate biting forces without breaking too easily. In the present tests, the reference cement gave the highest mean value, about 85 MPa. The cements containing samarium were somewhat lower, with mean values of around 69 MPa for Sm0.5, 70 MPa for Sm1 and 70 MPa for Sm5. These results are below 100 MPa, which is often used as a useful reference value for dental applications. Even so, this does not necessarily mean that the modified materials performed poorly.

The reason is that the samples tested here were not commercial restorative cements, but experimental materials prepared in the laboratory. They were experimental formulations prepared in the laboratory using synthesized glass powders. Therefore, their absolute strength values should not be directly compared with fully optimised commercial glass-ionomer cements. Commercial systems are usually developed through extensive adjustment of the glass composition, particle size distribution, powder/liquid ratio, additives, mixing procedure and maturation conditions. In the present study, the same powder/liquid ratio was intentionally used for all groups, so that the observed differences could be mainly associated with the Sm_2O_3 content rather than with changes in cement consistency.

In this context, the compressive strength results should be interpreted primarily in terms of general trends rather than as final application-ready performance. The statistical analysis showed no significant differences between the reference cement and the Sm-modified groups, with ANOVA giving *p* = 0.079. This indicates that samarium addition had only a limited influence on the bulk compressive strength of the experimental cements. Although the mean values were lower for all Sm-containing groups, this decrease was not statistically significant. Therefore, Sm_2O_3 cannot be considered a factor that strongly deteriorated the load-bearing capacity of the material within the investigated concentration range.

This observation is important because many functional modifications of glass-ionomer cements improve selected biological, antibacterial or surface-related properties at the cost of mechanical performance. In this case, samarium did not produce a statistically significant reduction in compressive strength, even at the highest concentration of 5 mol% Sm_2O_3. This suggests that samarium may be regarded as a relatively neutral modifier in terms of bulk compression behaviour. However, the fact that all values remained below 100 MPa also shows that the base laboratory formulation still requires further optimisation before it can be considered suitable for clinically demanding dental applications.

The lower absolute strength values may result not only from the presence of samarium, but also from the experimental nature of the cement system. The literature review included in the thesis indicates that the mechanical behaviour of glass-ionomer cements depends strongly on microstructure, particle size, the bond between glass particles and the polymer matrix, voids, storage conditions and the powder/liquid ratio. In particular, a higher powder/liquid ratio is generally associated with higher mechanical resistance. Therefore, further improvement of compressive strength may be possible by modifying the powder/liquid ratio, refining powder processing, improving mixing homogeneity and reducing porosity. This is especially relevant because, in this study, the powder/liquid ratio was kept constant to isolate the effect of Sm_2O_3.

In contrast to compressive strength, microhardness was strongly affected by samarium modification. The statistical analysis showed a significant effect of material composition on microhardness, with Welch-corrected ANOVA giving *p* < 0.001. Post-hoc analysis indicated statistically significant differences between all analysed groups. The lowest microhardness was observed for Sm0.5, followed by Sm1 and the reference material, whereas the highest value was obtained for Sm5. This indicates that the effect of samarium on surface mechanical properties was not linear.

The non-linear microhardness trend may be related to the dual role of Sm_2O_3 in the glass-ionomer system. At lower concentrations, samarium incorporation may disturb the fluoroaluminosilicate glass network or locally interfere with the acid–base reaction, resulting in lower surface hardness. At the highest concentration, however, the presence of Sm-containing phases or a modified glass-derived structure may increase local surface stiffness. This interpretation is consistent with the general mechanism of GIC setting, where the acid attacks the glass particles, releases metal ions and forms cross-linked polyacrylate structures responsible for cement hardening. The final material contains both reacted matrix regions and unreacted glass particles that act as reinforcement.

From an application point of view, the increase in microhardness observed for Sm5 may be beneficial. Surface hardness is related to resistance to indentation, surface degradation and, indirectly, wear behaviour. Since limited wear resistance is one of the known drawbacks of conventional glass-ionomer cements, the increase in hardness in the Sm5 group suggests that this composition may offer improved surface durability. However, this potential advantage should be considered together with the roughness results, because a harder surface is not necessarily more favourable if it also becomes rougher.

Surface roughness analysis showed that Sm_2O_3 also influenced surface topography. After the removal of one extreme outlier, the roughness data were analysed again. The Kruskal-Wallis test showed an overall difference in Ra values among the groups, $H(3, N = 19) = 8.56$, $p = 0.031$. The median test gave a similar result, $\chi^2(3) = 9.37$, $p = 0.025$. The Sm5 group had the highest mean rank, while the reference material, Sm0.5 and Sm1 showed lower and more similar roughness values. This suggests that the highest Sm_2O_3 content may have increased surface roughness.

However, the pairwise post-hoc comparisons did not show statistically significant differences between individual groups after correction. The closest comparison was between Sm1 and Sm5, with $p = 0.058$. This value is close to the 0.05 limit, but it is still not statistically significant. For this reason, the roughness results should be discussed carefully. The global tests indicate that material composition had an effect on R_a values, but the pairwise results only suggest a tendency towards higher roughness in the Sm5 group.

The simultaneous increase in microhardness and tendency towards higher roughness in Sm5 may indicate that the highest samarium concentration modified the near-surface structure of the cement. This could be associated with changes in glass particle reactivity, matrix formation, particle–matrix bonding, local hardness contrast between phases or polishing behaviour. In glass-ionomer cements, surface quality is closely linked to the distribution of glass particles and the surrounding polysalt matrix. If Sm_2O_3 changes the dissolution behaviour of the glass phase or the homogeneity of the hardened cement, the resulting surface may become harder but less smooth.

This point is particularly important for dental applications. A smooth surface is normally preferred in dental materials because it can reduce plaque retention, decrease mechanical irritation and give a better aesthetic result. For this reason, the tendency towards higher roughness in Sm5 may be a limitation, even though this group showed the best microhardness result. Sm0.5 and Sm1 behaved in a more moderate way. They did not improve hardness and did not produce a significant improvement in compressive strength, but they also did not show such a clear tendency towards higher roughness. This suggests that lower samarium contents may have a smaller effect on surface topography, whereas the highest concentration seems to modify the surface more strongly.

Overall, the results indicate that samarium mainly influenced surface-related properties rather than bulk compressive strength. The lack of statistically significant differences in compressive strength indicates that Sm_2O_3 does not strongly compromise the structural integrity of the experimental cements. At the same time, the significant changes in microhardness and the global differences in roughness show that samarium acts as an active modifier of the near-surface region. This distinction is important because many clinically relevant phenomena, including wear, bacterial adhesion, ion release, fluoride release and interaction with oral tissues, are surface-dependent.

Considering the expected biological benefits of samarium-containing biomaterials, the obtained results are promising but require balanced interpretation. Samarium-containing systems have been reported as relevant for dental and bone-related applications due to their biocompatibility, antibacterial potential, radiopacity, apatite-forming ability and possible contribution to mineralized tissue regeneration. However, in the present experimental GIC system, these potential advantages must be balanced against the observed mechanical and surface trends. Sm5 appears to be the most effective composition in terms of increasing microhardness, but it may also increase surface roughness. Sm0.5 and Sm1 seem less favourable from the hardness point of view, but their surface roughness remained closer to the reference material.

Therefore, the functional usefulness of samarium-modified glass-ionomer cements depends on the intended performance priority. If the aim is to obtain a stronger surface and potentially combine it with biological activity, Sm5 appears to be the most promising formulation. However, if surface smoothness is critical, the increased Ra tendency in Sm5 should be considered a limitation requiring further optimisation. Such optimisation could include increasing the powder/liquid ratio, modifying particle size distribution, improving powder homogenisation, adjusting the mixing procedure or refining the polishing protocol.

In conclusion, the investigated experimental cements did not reach the 100 MPa compressive strength level considered desirable for dental applications. However, this limitation should be attributed primarily to the early-stage, laboratory character of the formulations rather than to samarium addition alone. The most important trend is that Sm_2O_3 had only a limited and statistically non-significant effect on compressive strength, while its influence on surface properties was much more pronounced. Among the analysed compositions, Sm5 showed the highest microhardness and the strongest tendency towards increased roughness. This suggests that 5 mol% Sm_2O_3 may be the most active surface-modifying concentration, but further formulation optimisation is necessary to improve compressive strength and control surface roughness before considering practical dental application.

7. Conclusion

Based on the study carried out, the following conclusions have been drawn:

1) The experimental results showed that glass-ionomer cements containing samarium oxide can be successfully manufactured. SEM observations confirmed that these materials presented the general microstructural features expected in typical glass-ionomer cements. The main feature is that it is glass particles embedded in the cement matrix. Therefore, the incorporation of Sm_2O_3 did not prevent the formation of a glass-ionomer cement structure.
2) Regarding compressive strength, the incorporation of samarium oxide did not produce a significant improvement. The Sm-containing groups showed slightly lower mean values than the reference cement. However, this decrease was not statistically significant. Consequently, samarium can be considered a neutral modifier in terms of bulk compressive behaviour within the percentage range of Sm used.
3) Finally, the formulation still needs further optimisation before possible dental application. The Sm5 group showed a clear improvement in microhardness, which is positive for surface mechanical resistance. However, this group also showed a higher surface roughness, which is not desirable in dental materials. Therefore, future work should focus on keeping the hardness improvement while reducing roughness and improving the general mechanical performance.

REFERENCES

[1] S. Sidhu and J. Nicholson, “A Review of Glass-Ionomer Cements for Clinical Dentistry,” *J. Funct. Biomater.*, vol. 7, no. 3, p. 16, Jun. 2016, doi: 10.3390/jfb7030016.

[2] M. S. Baig and G. J. P. Fleming, “Conventional glass-ionomer materials: A review of the developments in glass powder, polyacid liquid and the strategies of reinforcement,” Aug. 01, 2015, *Elsevier Ltd*. doi: 10.1016/j.jdent.2015.04.004.

[3] A. Saatchifard, N. Nezafati, and S. Hesaraki, “Strategic approaches for enhancing the bioactivity of glass ionomer cement: A mechanistic and clinical perspective in terms of structural and surface modifications,” Dec. 01, 2025, *Elsevier Ltd*. doi: 10.1016/j.jdent.2025.106126.

[4] J. W. Nicholson, “Chemistry of glass-ionomer cements: a review,” 1998.

[5] J. W. Nicholson, “Maturation processes in glass-ionomer dental cements,” *Acta Biomater. Odontol. Scand.*, vol. 4, no. 1, pp. 63–71, Jan. 2018, doi: 10.1080/23337931.2018.1497492.

[6] S. Bahammam, D. Nathanson, and Y. Fan, “Evaluating the Mechanical Properties of Restorative Glass Ionomers Cements,” *Int. Dent. J.*, vol. 72, no. 6, pp. 859–865, Dec. 2022, doi: 10.1016/j.identj.2022.06.016.

[7] D. Xie, W. A. Brantley, B. M. Culbertson, and G. Wang, “Mechanical properties and microstructures of glass-ionomer cements.” [Online]. Available: www.elsevier.com/locate/dental

[8] D. Dionysopoulos, O. Gerasimidou, and C. Papadopoulos, “Modifications of Glass Ionomer Cements Using Nanotechnology: Recent Advances,” *Recent Prog. Mater.*, vol. 04, no. 02, pp. 1–22, Jun. 2022, doi: 10.21926/rpm.2202011.

[9] J. W. Nicholson, S. K. Sidhu, and B. Czarnecka, “Enhancing the mechanical properties of glass-ionomer dental cements: A review,” Jun. 01, 2020, *MDPI AG*. doi: 10.3390/ma13112510.

[10] N. Khosravi, A. Moradi, F. Sharifianjazi, K. Tavamaishvili, A. Bakhtiari, and A. Mohammadi, “Review of samarium-containing bioactive glasses: biomedical applications,” *Journal of Composites and Compounds*, vol. 7, no. 22, Feb. 2025, doi: 10.61186/jcc.7.1.3.

[11] T. Tüzüner, A. Dimkov, and J. W. Nicholson, “The effect of antimicrobial additives on the properties of dental glass-ionomer cements: a review,” *Acta Biomater. Odontol. Scand.*, vol. 5, no. 1, pp. 9–21, Jan. 2019, doi: 10.1080/23337931.2018.1539623.

[12] T. De Caluwé *et al.*, “Addition of bioactive glass to glass ionomer cements: Effect on the physico-chemical properties and biocompatibility,” *Dental Materials*, vol. 33, no. 4, pp. e186–e203, Apr. 2017, doi: 10.1016/j.dental.2017.01.007.

[13] M. Balas *et al.*, “Biocompatibility and Osteogenic Activity of Samarium-Doped Hydroxyapatite—Biomimetic Nanoceramics for Bone Regeneration Applications,” *Biomimetics*, vol. 9, no. 6, Jun. 2024, doi: 10.3390/biomimetics9060309.

[14] A. N. Saud, E. Koç, O. Özdemir, Y. Cetin, and Y. Yildizhan, “Development and Evaluation of Bioactive Glass Ceramic-Based Endodontic Cement: A Comprehensive Study on Bioactivity, Biocompatibility, Mechanical Properties, Radiopacity, and Antimicrobial Activity,” *J. Mater. Eng. Perform.*, vol. 34, no. 13, pp. 12750–12767, Jul. 2025, doi: 10.1007/s11665-024-10098-1.

[15] I. C. Nica *et al.*, “Biocompatibility and antibiofilm properties of samarium doped hydroxyapatite coatings: An in vitro study,” *Coatings*, vol. 11, no. 10, Oct. 2021, doi: 10.3390/coatings11101185.

[16] C. P. Harini, R. Gosala, and R. Ramadoss, “Samarium-doped hydroxyapatite: An effective approach for enhancing biomineralization in dental caries management,” *J. Oral Biol. Craniofac. Res.*, vol. 15, no. 6, pp. 1204–1209, Nov. 2025, doi: 10.1016/j.jobcr.2025.07.026.

[17] T. De Caluwé, C. W. J. Vercruysse, S. Fraeyman, and R. M. H. Verbeeck, “The influence of particle size and fluorine content of aluminosilicate glass on the glass ionomer cement properties,” *Dental Materials*, vol. 30, no. 9, pp. 1029–1038, 2014, doi: 10.1016/j.dental.2014.06.003.

[18] H. S. Ching, N. Luddin, T. P. Kannan, I. Ab Rahman, and N. R. N. Abdul Ghani, “Modification of glass ionomer cements on their physical-mechanical and antimicrobial properties,” Nov. 01, 2018, *Blackwell Publishing Ltd*. doi: 10.1111/jerd.12413.

[19] J. Makanjuola and S. Deb, “Chemically Activated Glass-Ionomer Cements as Bioactive Materials in Dentistry: A Review,” Mar. 01, 2023, *MDPI*. doi: 10.3390/prosthesis5010024.

[20] S. Najeeb *et al.*, “Modifications in glass ionomer cements: Nano-sized fillers and bioactive nanoceramics,” Jul. 14, 2016, *MDPI AG*. doi: 10.3390/ijms17071134.

[21] S. Mahmood, S. Shahid, and R. Billington, “DO GLASS IONOMER CEMENTS MATURE APPRECIABLY? A CRITICAL REVIEW.”

[22] M. Łępicka *et al.*, “The influence of hydrothermal fatigue on the clinically relevant functional properties of conventional glass-ionomer cements,” *Sci. Rep.*, vol. 13, no. 1, Dec. 2023, doi: 10.1038/s41598-023-35880-4.

[23] L. Al-Taee, S. Deb, and A. Banerjee, “An in vitro assessment of the physical properties of manually- mixed and encapsulated glass-ionomer cements,” *BDJ Open*, vol. 6, no. 1, Dec. 2020, doi: 10.1038/s41405-020-0040-x.

[24] J. W. Nicholson, “Adhesion of glass-ionomer cements to teeth: A review,” 2016, *Elsevier Ltd*. doi: 10.1016/j.ijadhadh.2016.03.012.

[25] K. Nakajo, S. Imazato, Y. Takahashi, W. Kiba, S. Ebisu, and N. Takahashi, "Fluoride released from glass-ionomer cement is responsible to inhibit the acid production of caries-related oral streptococci," *Dental Materials*, vol. 25, no. 6, pp. 703–708, Jun. 2009, doi: 10.1016/j.dental.2008.10.014.

[26] M. Łukomska-Szymańska *et al.*, "Antibacterial properties of calcium fluoride-based composite materials: In vitro study," *Biomed Res. Int.*, vol. 2016, 2016, doi: 10.1155/2016/1048320.

[27] M. A. A. De Bruyne and R. J. G. De Moor, "The use of glass ionomer cements in both conventional and surgical endodontics," 2004, *Blackwell Publishing Ltd*. doi: 10.1111/j.0143-2885.2004.00769.x.

[28] S. Hoshika *et al.*, "Effect of conditioning and 1 year aging on the bond strength and interfacial morphology of glass-ionomer cement bonded to dentin," *Dental Materials*, vol. 37, no. 1, pp. 106–112, Jan. 2021, doi: 10.1016/j.dental.2020.10.016.

[29] S. Mostafa Mousavinasab and I. Meyers, "Fluoride Release by Glass Ionomer Cements, Compomer and Giomer," 2009.

[30] E. Łuczaj-Cepowicz, G. Marczuk-Kolada, A. Zalewska, M. Pawińska, and K. Leszczyńska, "Antibacterial activity of selected glass ionomer cements Aktywność przeciwbakteryjna wybranych cementów szkłojonomerowych," *Postepy Hig Med Dosw*, vol. 68, p. 68, 2014, [Online]. Available: www.phmd.pl

[31] K. J. Chun, H. H. Choi, and J. Y. Lee, "Comparison of mechanical property and role between enamel and dentin in the human teeth," *J. Dent. Biomech.*, vol. 5, no. 1, pp. 1–7, 2014, doi: 10.1177/1758736014520809.

[32] D. Zaytsev, "Mechanical properties of human enamel under compression: On the feature of calculations," *Materials Science and Engineering C*, vol. 62, pp. 518–523, May 2016, doi: 10.1016/j.msec.2016.02.016.

[33] S. K. Sidhu, "Clinical evaluations of resin-modified glass-ionomer restorations," *Dental Materials*, vol. 26, no. 1, pp. 7–12, Jan. 2010, doi: 10.1016/j.dental.2009.08.015.

[34] M. S. Gale and B. W. Darvell, "Thermal cycling procedures for laboratory testing of dental restorations."

[35] D. Natarajan, Z. Ye, L. Wang, L. Ge, and J. L. Pathak, "Rare earth smart nanomaterials for bone tissue engineering and implantology: Advances, challenges, and prospects," Jan. 01, 2022, *John Wiley and Sons Inc*. doi: 10.1002/btm2.10262.

[36] Q. Zhang, S. O'brien, and J. Grimm, "Biomedical applications of lanthanide nanomaterials, for imaging, sensing and therapy," 2022, *Ivyspring International Publisher*. doi: 10.7150/NTNO.65530.

[37] K. Staszak, M. Rzelewska-Piekut, and M. Regel-Rosocka, "Role of noble and rare earth metals in bioactive materials for medical applications in tissue

engineering," *RSC Adv.*, vol. 15, no. 48, pp. 40709–40729, 2025, doi: 10.1039/d5ra04036a.

[38] M. del C. De Lama-Odría, L. J. del Valle, and J. Puiggalí, "Lanthanides-Substituted Hydroxyapatite for Biomedical Applications," Feb. 01, 2023, *MDPI*. doi: 10.3390/ijms24043446.

[39] C. P. Harini, R. Gosala, and R. Ramadoss, "Samarium-doped hydroxyapatite: An effective approach for enhancing biomineralization in dental caries management," *J. Oral Biol. Craniofac. Res.*, vol. 15, no. 6, pp. 1204–1209, Nov. 2025, doi: 10.1016/j.jobcr.2025.07.026.

[40] A. Sajjad, W. Wan Bakar, D. Mohamad, and T. Kannan, "Various recent reinforcement phase incorporations and modifications in glass ionomer powder compositions: A comprehensive review," Jul. 01, 2018, *Wolters Kluwer Medknow Publications*. doi: 10.4103/jioh.jioh_160_18.

[41] H. A. Abo-Mosallam and H. C. Park, "Crystallization characteristics of La2O3-containing glasses for glass ionomer cement," *Mater. Lett.*, vol. 72, pp. 137–140, Apr. 2012, doi: 10.1016/J.MATLET.2011.12.111.

[42] G.U.N.T. Gerätebau GmbH, "Mechanical testing methods." Accessed: May 23, 2026. [Online]. Available: https://www.gunt.de/images/download/mechanical_testing_methods_english.pdf

[43] Instron, "What is Compression Testing?" Accessed: May 23, 2026. [Online]. Available: https://www.instron.com/en/resources/test-types/compression-test/

[44] Richard Gedney, "Compression Testing Fundamentals," *Quality Magazine*, Oct. 08, 2024. Accessed: May 23, 2026. [Online]. Available: https://www.qualitymag.com/articles/98320-compression-testing-fundamentals

[45] Serope Kalpakjian; Steven R. Schmid, "Manufacturing Engineering and Technology," 6th ed., Upper Saddle River, NJ, USA: Prentice Hall, 2010, ch. 2, pp. 56–87. Accessed: May 23, 2026. [Online]. Available: https://www.pearsonhighered.com/assets/samplechapter/0/1/3/6/0136081681.pdf

[46] Inc. MechaniCalc, "Mechanical Properties of Materials." Accessed: May 23, 2026. [Online]. Available: https://mechanicalc.com/reference/mechanical-properties-of-materials

[47] Thermo Fisher Scientific, "Desktop SEM Sample Preparation Guide," Desktop SEM Sample Preparation Guide. Accessed: May 23, 2026. [Online]. Available: https://documents.thermofisher.com/TFS-Assets/MSD/Application-Notes/desktop-sem-sample-preparation-guide-BR0098.pdf

[48] "Grażyna GILEWSKA PRZYDATNOŚĆ RÓŻNYCH TECHNIK OBRAZOWANIA STRUKTUR BIOLOGICZNYCH WYKORZYSTUJĄCYCH ELEKTRONOWY MIKROSKOP SKANINGOWY *)."

[49] KEYENCE Corporation, "Non-Contact Surface Roughness/Profile Measuring Instruments," KEYENCE America. Accessed: May 23, 2026. [Online]. Available:

https://www.keyence.com/ss/products/microscope/roughness/equipment/non-contact-instruments.jsp

[50] Norsheila Binti Buyamin; Zanidah Binti Ithnin; Rawaida Binti Muhammad, *Material Testing*. Ipoh, Perak, Malaysia: Politeknik Ungku Omar, 2023. Accessed: May 23, 2026. [Online]. Available: https://repositori.mypolycc.edu.my/bitstream/123456789/2000/1/eBook_Material%20Testing_DKM_JKM.pdf